# Category

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# 1 Categories, functors and natural transformations

**Definition 1.1.** A **category** $\mathcal{C}$ consists of

1. a class $\mathrm{Ob}(\mathcal{C})$ of **objects**

2. for any pair of objects $A, B$, a class $\mathrm{Hom}_{\mathcal{C}}(A, B)$ of all **morphisms** from $A$ to $B$, where $A$ is called the **source**, or **domain**, and $B$ is called the **target**, or **codomain**, of morphisms.

3. for each triple of objects $A, B, C$, a function $\circ : \mathrm{Hom}_{\mathcal{C}}(A, B) \times \mathrm{Hom}_{\mathcal{C}}(B, C) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$ that assigns each appropriate pair of morphisms $f, g$ to their **composite** morphism $g \circ f$.

4. for each object $A$, an (unique) element $\mathrm{id}_A$ of $\mathrm{Hom}_{\mathcal{C}}(A, A)$, the **identity** morphism.

such that the following properties hold:

- the composite is **associative**: for each quadruple of objects $A, B, C, D$, if $f$ in $\mathrm{Hom}_{\mathcal{C}}(A, B)$, $g$ in $\mathrm{Hom}_{\mathcal{C}}(B, C)$ and $h$ in $\mathrm{Hom}_{\mathcal{C}}(C, D)$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f$$

- the composite is **unital**: for any pair of objects $A, B$, if $f$ in $\mathrm{Hom}_{\mathcal{C}}(A, B)$, we have

$$f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$$

**Notation 1.2.** There are some common notations used in the context of the category.

1. We write $A \in \mathrm{Ob}(\mathcal{C})$, or simply $A \in \mathcal{C}$ to mean that $A$ is an object of $\mathcal{C}$.
2. We write $A \xrightarrow{f} B$ to mean an morphism from $A$ to $B$, i.e, an element of $\mathrm{Hom}_{\mathcal{C}}(A, B)$. Also, we write $\left(A \xrightarrow{f} B\right) \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ or $f \in \mathrm{Hom}(\mathcal{C})(A, B)$, and write $f \in \mathcal{C}$ to mean $f$ is a morphism in $\mathcal{C}$.
3. If the context is clear, we write $\mathrm{Hom}(A, B)$ rather than $\mathrm{Hom}_{\mathcal{C}}(A, B)$.
4. We write $\mathrm{Mor}(\mathcal{C})$ to mean the "disjoint union" $\coprod_{A \in \mathrm{Ob}(\mathcal{C})} \coprod_{B \in \mathrm{Ob}(\mathcal{C})} \mathrm{Hom}_{\mathcal{C}}(A, B)$, that is, each **hom-class** is regarded to be disjoint.
5. We write $\mathrm{End}(A)$ to mean $\mathrm{Hom}(A, A)$. An **endomorphism** is a morphism whose domain and codomain coincide. Clearly, $\mathrm{End}(A)$ forms a monoid under composition when hom-classes are really sets.

**Definition 1.3.**

1. A category $\mathcal{C}$ is **small** if $\mathrm{Ob}(\mathcal{C})$ and $\mathrm{Mor}(\mathcal{C})$ are sets. Otherwise, it's **large**.
2. A category $\mathcal{C}$ is **locally small** if for each pair of objects $A, B$, $\mathrm{Hom}(A, B)$ is a set, called a **homset**.

**Example 1.4.**

1. (**Set**): the category of sets. Sets are as objects, and maps are as morphisms.
2. (**Grp**): the category of groups. Groups as objects, group homomorphisms as morphisms.
3. (**AbGrp**): the category of abelian groups.
4. (**Ring**): the category of rings with identity. Rings with identity as objects, ring homomorphisms preserving identity as morphisms.

5. (**Rng**): the category of rings. Rings as objects, ring homomorphisms as morphisms.

6. ($R$-**Mod**): the category of $R$-modules.

7. ($\mathbf{Vec}_k$): the category of $k$-vector space.

8. (**Top**): the category of topological spaces. Spaces as objects, continuous maps as morphisms.

9. (**Top.**): the category of pointed topological spaces. Pointed spaces as objects, pointed continuous maps as morphisms.

10. (**Top**($X$)): the category of open sets in $X$. Open sets are as objects and

$$\mathrm{Hom}(A, B) = \begin{cases} \{\iota : A \hookrightarrow B\} & \text{, if } A \subseteq B \\ \varnothing & \text{, otherwise} \end{cases}$$

11. In general, for each set $S$, we may form a category as follows: the object is the power $\mathcal{P}(S)$, and

$$\mathrm{Hom}(A, B) = \begin{cases} \{\iota : A \hookrightarrow B\} & \text{, if } A \subseteq B \\ \varnothing & \text{, otherwise} \end{cases}$$

12. (**Rel**): the category of relations. Sets as objects, binary relations as morphisms. The composition of two relation $R : A \to B$ and $S : B \to C$ are defined as

$$(a, c) \in S \circ R \Leftrightarrow (a, b) \in R \wedge (b, c) \in S \text{ for some } b \in B$$

13. Let $S$ be as set and $R$ be a reflexive transitive binary relation. Then we may form a category: objects as the elements of $S$, and if $a, b \in S$, then

$$\mathrm{Hom}(a, b) = \begin{cases} \{(a, b)\} & \text{, if } (a, b) \in R \\ \varnothing & \text{, otherwise} \end{cases}$$

The composition can be defined in a natural way, by virtues of transitivity. The identity morphism $\mathrm{id}_a$ for each element is just $(a, a)$, by virtues of reflexivity.

14. Let $\mathcal{C}$ be a category. Its **arrow category** $\mathrm{Arr}(\mathcal{C})$ is the category such that:

- a object $a$ is a morphism $a : A \to B$ in $\mathcal{C}$.
- a morphism $f : a \to a'$ of is a commutative square

$$\begin{tikzcd} A \arrow[r, "f_1"] \arrow[d, "a"'] & A' \arrow[d, "a'"] \\ B \arrow[r, "f_2"] & B' \end{tikzcd}$$

in $\mathcal{C}$.

- the composition of morphisms is given by placing commutative squares side by side.

15. Let $G$ be a group. We may form a category as follows: the only object is $G$ and $\mathrm{Hom}(G, G) = G$. The composition of two morphisms $f, g$ is the product $gf$ in the group. The identity morphism is the identity element of $G$.

16. Let $\mathcal{C}$ be a category. The **opposite category** $\mathcal{C}^{\mathrm{op}}$ the category such that

    - $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$
    - for each pair of objects $A, B$, we have $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$

    In other words, all the arrows are reversed in the opposite category.

## 1.1 Morphisms

**Definition 1.5.** Let $\mathcal{C}$ be a category and $A \xrightarrow{f} B$ be a morphism.

1. $f$ is an **isomorphism** if there's a morphism $B \xrightarrow{g} A$ such that $f \circ g = \mathrm{id}_B$ and $g \circ f = \mathrm{id}_A$. In this case, we say $g$ is the **inverse** of $f$, and two objects $A, B$ are isomorphic, denoted by $A \cong B$.

2. $f$ is an **monomorphism** if for each pair of parallel morphisms $C \underset{g_2}{\overset{g_1}{\rightrightarrows}} A$ ,

$$f \circ g_1 = f \circ g_2 \Rightarrow g_1 = g_2$$

3. $f$ is an **epimorphism** if for each pair of parallel morphisms $B \underset{g_2}{\overset{g_1}{\rightrightarrows}} C$ ,

$$g_1 \circ f = g_2 \circ f \Rightarrow g_1 = g_2$$

- One may check if an inverse of $f$ exists, then it's unique. This justifies our notation $f^{-1}$.
- An **automorphism** is an endomorphism that is also an isomorphism; by $\mathrm{Aut}(A)$ we denote the collection of all automorphisms on $A$. It can be easily verified that $\mathrm{Aut}(A)$ forms a group under composition when $\mathcal{C}$ is locally small.

**Property 1.6.** Let $\mathcal{C}$ be a category and $A \xrightarrow{f} B \xrightarrow{g} C$ be morphisms. Then:

1. If $f$ and $g$ are isomorphisms (resp. monomorphisms, epimorphisms), then so is $g \circ f$.
2. If $g \circ f$ is a monomorphism (resp. epimorphism), then so is $f$ (resp. $g$).
3. If $g \circ f$ is an isomorphism and $g$ is a monomorphism, then both $f$ and $g$ are isomorphisms.
4. If $f$ is an isomorphism, then it's also a monomorphism and an epimorphism.

**Example 1.7.** In general, the converse of 4. is false, however. For instance, consider the inclusion map $\mathbb{Z} \to \mathbb{Q}$ in the category of rings.

However, in the category of commutative rings with identity, the monomorphisms are exactly injective ring homomorphisms preserving identity. Indeed, let $R, R'$ be a commutative ring with identity and $f : R \to R'$ be a ring homomorphism preserving identity. For each element $r \in R$, consider the ring homomorphism

$$\begin{aligned} \overline{r} : \mathbb{Z}[x] &\longrightarrow R \\ p &\longmapsto p(r) \end{aligned}$$

Now if $f(r) = f(s)$, this means $(f \circ \overline{r})(x) = (f \circ \overline{s})(x)$. Note that a ring homomorphism from $\mathbb{Z}[x]$ is determined by its action on 1 and $x$, and this implies $f \circ \overline{r} = f \circ \overline{s}$, thus $\overline{r} = \overline{s}$, i.e, $r = s$.

**Example 1.8.** Consider the category of connected pointed spaces, in which morphisms are base-point preserving continuous map, i.e, pointed maps. Let $\pi : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ be a covering map and $f : (Y, y_0) \to (X, x_0)$ be a pointed map that admits a lifting $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x_0})$. Then $\tilde{f}$ is uniquely determined. This precisely means that $\pi$ is a monomorphism.

**Example 1.9.** Though epimorphisms in the category of topological spaces are precisely continuous surjections, it's not the case in the category of Hausdorff spaces. Instead, a continuous map $f : X \to Y$ is an epimorphism if and only if $f(X)$ is dense in $Y$. The if part is obvious. Suppose now that $f : X \to Y$ is an epimorphism. First note that if $Y$ is nonempty, then so is $X$. Indeed, that $X$ is empty leads to a contradiction that two canonical inclusions $Y \to Y \coprod Y$ are the same. Consider the canonical projection $\pi : Y \to Y/\overline{f(X)}$. Note that a quotient of Hausdorff spaces by a closed subspace is Hausdorff. Consider as well the constant mapping $c : Y \to Y/\overline{f(X)}$ on the equivalence class of the elements of $\overline{f(X)}$. Clearly, $c \circ f = \pi \circ f$, thus $c = \pi$, which is exactly the equality $\overline{f(X)} = Y$.

Although the converse of 4. does not hold in general, there's still a partial converse:

**Property 1.10.** A monomorphism $f : A \to B$ having a right inverse $g : B \to A$, i.e, $f \circ g = \mathrm{id}_G$, is an isomorphism. Dually, an epimorphism having a left inverse is an isomorphism.

**Definition 1.11.** Let $\mathcal{C}$ be a category. A **subcategory** of $\mathcal{C}$ is a category $\mathcal{C}'$ such that $\mathrm{Ob}(\mathcal{C}') \subseteq \mathrm{Ob}(\mathcal{C})$ and for each pair $A, B \in \mathrm{Ob}(\mathcal{C}')$, $\mathrm{Hom}_{\mathcal{C}'}(A, B) \subseteq \mathrm{Hom}_{\mathcal{C}}(A, B)$.

- A subcategory $\mathcal{C}'$ of $\mathcal{C}$ is **full** if $\mathrm{Hom}_{\mathcal{C}'}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$ for each pair $A, B \in \mathrm{Ob}(\mathcal{C}')$.

**Example 1.12.** Let $\mathcal{C}$ be a category. The category $\mathcal{C}_{\mathrm{m}}$, whose objects are the same as $\mathcal{C}$ and $\mathrm{Hom}_{\mathcal{C}_{\mathrm{m}}}(A, B)$ are monomorphisms in $\mathrm{Hom}_{\mathcal{C}}(A, B)$, is a non-full subcategory of $\mathcal{C}$. The statement remains true when monomorphism is replaced by epimorphism.

## 1.2 Functors

**Definition 1.13.** Let $\mathcal{C}$ and $\mathcal{C}'$ be category. A **covariant functor** $\mathcal{F} : \mathcal{C} \to \mathcal{C}'$ assigning

1. every object $A$ in $\mathcal{C}$ to $\mathcal{F}(A)$ an object in $\mathcal{C}'$
2. every morphism $A \xrightarrow{f} B$ to a morphism $\mathcal{F}(A) \xrightarrow{\mathcal{F}(f)} (\mathcal{F})B$ in $\mathcal{C}'$ such that
   (a) for each object $A$ in $\mathcal{C}$, $\mathcal{F}(\mathrm{id}_A) = \mathrm{id}_{\mathcal{F}(A)}$
   (b) for any composition $A \xrightarrow{f} B \xrightarrow{g} C$, we have $\mathcal{F}A \xrightarrow{\mathcal{F}f} \mathcal{F}B \xrightarrow{\mathcal{F}g} \mathcal{F}C$, that is, $\mathcal{F}(gf) = \mathcal{F}(g)\mathcal{F}(f)$.

**Definition 1.14.** Let $\mathcal{C}$ and $\mathcal{C}'$ be category. A **contravariant functor** $\mathcal{F} : \mathcal{C} \to \mathcal{C}'$ assigning

1. every object $A$ in $\mathcal{C}$ to $\mathcal{F}(A)$ an object in $\mathcal{C}'$
2. every morphism $A \xrightarrow{f} B$ to a morphism $\mathcal{F}(B) \xrightarrow{\mathcal{F}(f)} (\mathcal{F})A$ in $\mathcal{C}'$ such that
   (a) for each object $A$ in $\mathcal{C}$, $\mathcal{F}(\mathrm{id}_A) = \mathrm{id}_{\mathcal{F}(A)}$
   (b) for any composition $A \xrightarrow{f} B \xrightarrow{g} C$, we have $\mathcal{F}C \xrightarrow{\mathcal{F}g} \mathcal{F}B \xrightarrow{\mathcal{F}f} \mathcal{F}A$, that is, $\mathcal{F}(gf) = \mathcal{F}(f)\mathcal{F}(g)$.

- Every contravariant functor $\mathcal{C} \to \mathcal{C}'$ can be viewed as a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{C}'$.

**Definition 1.15.** Let $\mathcal{F} : \mathcal{C} \to \mathcal{C}'$ be a covariant functor between locally small categories.

1. $\mathcal{F}$ is **faithful** (resp. **full**) if
$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}'}(\mathcal{F}X, \mathcal{F}Y)$$
   is injective (resp. surjective) for any pair of objects $X, Y$.
2. $\mathcal{F}$ is **fully faithful** if
$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}'}(\mathcal{F}X, \mathcal{F}Y)$$
   is bijective for any pair of objects $X, Y$.

**Example 1.16.**

1. Let $\mathcal{C}$ be a locally small category and $A$ be an object. Define the contravariant functor $h_A : \mathcal{C} \to (\mathbf{Set})$ by

   (a) for any $X \in \mathrm{Ob}(\mathcal{C})$, $h_A(X) := \mathrm{Hom}_{\mathcal{C}}(X, A)$

   (b) for any $X \xrightarrow{f} Y$,
   $$h_A(f) \in \mathrm{Hom}_{(\mathbf{Set})}(\mathrm{Hom}_{\mathcal{C}}(Y, A), \mathrm{Hom}_{\mathcal{C}}(X, A))$$
   defined by $h_A(f)\left(Y \xrightarrow{u} A\right) = \left(X \xrightarrow{u \circ f} A\right)$, where $u \circ f$ is the composition in $\mathcal{C}$.

2. Let $\mathcal{C}$ be a locally small category and $A$ be an object. Define the covariant functor ${}_Ah : \mathcal{C} \to (\mathbf{Set})$ by

   (a) for any $X \in \mathrm{Ob}(\mathcal{C})$, ${}_Ah(X) := \mathrm{Hom}_{\mathcal{C}}(A, X)$

   (b) for any $X \xrightarrow{f} Y$,
   $${}_Ah(f) \in \mathrm{Hom}_{(\mathbf{Set})}(\mathrm{Hom}_{\mathcal{C}}(A, X), \mathrm{Hom}_{\mathcal{C}}(A, Y))$$
   defined by ${}_Ah(f)\left(A \xrightarrow{u} X\right) = \left(A \xrightarrow{f \circ u} Y\right)$, where $f \circ u$ is the composition in $\mathcal{C}$.

3. Let $\mathcal{C}$ be a category. Then ${}^{\mathrm{op}} : \mathcal{C} \to \mathcal{C}^{\mathrm{op}}$ is a contravariant functor.

4. The fundamental group $\pi_1 : (\mathbf{Top.}) \to (\mathbf{Grp})$ is a covariant functor.

5. ${}^{\times} : (\mathbf{Ring}) \to (\mathbf{Grp})$ is a covariant functor assigning each ring to its group of units.

6. For each group $G$, define $G^{\mathrm{ab}} = G/[G, G]$, and for any morphism $f : G_1 \to G_2$, $f^{\mathrm{ab}}$ is defined canonically. Then ${}^{\mathrm{ab}} : (\mathbf{Grp}) \to (\mathbf{Grp})$ is a covariant functor, called the **abelianization functor**.

7. Let $E/F$ be a Galois extension and $G = \mathrm{Gal}(E/F)$. By Galois theory, we know there's a one-to-one inclusion-reversing correspondence of the intermediate field $F \subseteq K \subseteq E$ and the subgroup $G \geqslant H = \mathrm{Gal}(K/F) \geqslant 1$. If $F \subseteq A \subseteq B \subseteq E$, we have a natural inclusion $\mathrm{Gal}(B/E) \hookrightarrow \mathrm{Gal}(A/E)$, induced by the inclusion $A \hookrightarrow B$. Hence Gal is a contravariant functor from the category of the intermediate fields of $F \subseteq E$ and the category of subgroups of $G$.

8. Let $F : (\mathbf{Top}) \to (\mathbf{Set})$ be the functor which assigning each topological space to its underlying set. Then $F$ is a faithful covariant functor, called the **forgetful functor**.

9. Let $\mathcal{C}$ be a category. Then $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is a covariant functor leaving everything unchanged, called the **identity functor**.

10. Let $J$ an $\mathcal{C}$ be categories. A **diagram of type** $J$ is a covariant functor $F : J \to \mathcal{C}$. In this case, $J$ is called an **index category**. We say $F$ is a small (resp. finite) diagram if $J$ is small (resp. finite).

**Property 1.17.** Let $\mathcal{C}$ be a locally small category, $A$ an object of $\mathcal{C}$, and $X \xrightarrow{f} Y$ a morphism in $\mathcal{C}$. Then

1. $f$ is a monomorphism if and only if ${}_Ah(f) : \mathrm{Hom}_{\mathcal{C}}(A, X) \to \mathrm{Hom}_{\mathcal{C}}(A, Y)$ is injective.

2. $f$ is an epimorphism if and only if $h_A(f) : \mathrm{Hom}_{\mathcal{C}}(Y, A) \to \mathrm{Hom}_{\mathcal{C}}(X, A)$ is injective.

**Property 1.18.** Let $F : \mathcal{C} \to \mathcal{C}'$ be a faithful functor and let $X \xrightarrow{f} Y$ be a morphism in $\mathcal{C}$. Then if $F(f)$ is a monomorphism (resp. an epimorphism), then so is $f$. Furthermore, if $F$ is fully faithful, then $F(f)$ is an isomorphism implies $f$ is an isomorphism.

**Definition 1.19.** Let $\mathcal{C}, \mathcal{C}', \mathcal{C}''$ be categories and $F : \mathcal{C} \to \mathcal{C}'$, $G : \mathcal{C}' \to \mathcal{C}''$ be two covariant functors. Their **composition** $G \circ F : \mathcal{C} \to \mathcal{C}'$ is the covariant functor defined by

- $(G \circ F)(X) = G(F(X))$ for each object $X$ in $\mathcal{C}$.

- $(G \circ F)(f) = G(F(f))$ for each morphism $f$ in $\mathcal{C}$.

Similarly, if $F, G$ are contravariant, we may define $G \circ F$ canonically to be a covariant functor.

## 1.3 Natural transformation

**Definition 1.20.** Let $\mathcal{C}, \mathcal{C}'$ be categories and $F : \mathcal{C} \to \mathcal{C}'$, $G : \mathcal{C} \to \mathcal{C}'$ be covariant functors. A **natural transformation** $T : F \to G$ assigns to each object $A$ of $\mathcal{C}$ a morphism $T(A) : F(A) \to G(A)$ such that for any morphism $X \xrightarrow{f} Y$ in $\mathcal{C}$, the following diagram commutes in $\mathcal{C}'$:

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "T(X)"'] & F(Y) \arrow[d, "T(Y)"] \\ G(X) \arrow[r, "G(f)"'] & G(Y) \end{tikzcd}$$

- $F$ is called a **natural isomorphism** if for each object $A$ of $\mathcal{C}$, $T(A) : F(A) \to G(A)$ is an isomorphism in $\mathcal{C}'$.

- Let $F, C : \mathcal{C} \to \mathcal{C}'$ be functors. We say the (collection of) maps $T(X) : F(X) \to G(X)$ is **functorial in** $X$, or **natural in** $X$, if $T : F \to G$ is a natural transformation.

**Example 1.21.**

1. Let ${}^{\mathrm{ab}} : (\mathbf{Grp}) \to (\mathbf{Grp})$ be the abelianization functor. For any group $G$, define $\pi(G) : G \to G/[G,G]$ to be the projection. Then $\pi$ is an natural transformation from the identity functor to the abelianization functor. In other words, every homomorphism $f : G_1 \to G_2$ naturally induces a homomorphism $f^{\mathrm{ab}} : G_1^{\mathrm{ab}} \to G_2^{\mathrm{ab}}$ such that $\pi(G_2) \circ f = f^{\mathrm{ab}} \circ \pi(G_1)$.

2. Let $(\mathbf{CRng})$ be the category of commutative rings, and let $\mathrm{GL}_n : (\mathbf{CRng}) \to (\mathbf{Grp})$ be the functor sending each object $R$ to the group $\mathrm{GL}_n(R)$ of $n \times n$ invertible matrix; any ring homomorphism $R \xrightarrow{f} S$ defines $\mathrm{GL}_n(R) \xrightarrow{\mathrm{GL}_n(f)} \mathrm{GL}_n(S)$ entry by entry. det, the determinant, is the natural transformation from $\mathrm{GL}_n$ to ${}^{\times}$, i.e, for any ring homomorphism $R \xrightarrow{f} S$, the following diagram commutes:

$$\begin{tikzcd}
\mathrm{GL}_n(R) \arrow[r, "\mathrm{GL}_n(f)"] \arrow[d, "\det(R)"'] & \mathrm{GL}_n(S) \arrow[d, "\det(S)"] \\
R^{\times} \arrow[r, "f^{\times}"'] & S^{\times}
\end{tikzcd}$$

3. Let $\mathcal{C}$ be a small category $\mathcal{C}'$ a locally small category, and $F, G, H : \mathcal{C} \to \mathcal{C}'$ covariant functors. Suppose there are natural transformations $\nu : F \to G$ and $\mu : G \to H$. Then their **(vertical) composition** $\mu \circ \nu : F \to H$ can be defined canonically. Hence, we may form a category such that objects are covariant functors from $\mathcal{C}$ to $\mathcal{C}'$, and morphisms are natural transformations, called the **category of functors from $\mathcal{C}$ to $\mathcal{C}'$**, or simply **functor category**, and denoted by $[\mathcal{C}, \mathcal{C}']$, $\mathrm{Funct}(\mathcal{C}, \mathcal{C}')$ or $\mathcal{C}'^{\mathcal{C}}$. One can see $[\mathcal{C}, \mathcal{C}']$ is locally small. Also, it's easy to see that the isomorphisms in $[\mathcal{C}, \mathcal{C}']$ are exactly the natural isomorphisms.

   - One can also consider the case when $\mathcal{C}$ is not small. However, if $\mathcal{C}$ is not assumed to be small, then even if $\mathcal{C}'$ is locally small, $[\mathcal{C}, \mathcal{C}']$ may still fail to a locally small category: consider a category $\mathcal{C}$ whose objects form a proper class such that no morphism is between two distinct objects and each object has a cyclic endomorphism group of order 2. Let $F = G = \mathrm{id}_{\mathcal{C}}$ be the identity functor. Each natural transformation is determined by the assignment to each object an endomorphism, which has two choices. Clearly all such natural transformations form a proper class rather than a set.
   - In the case $\mathcal{C}' = (\mathbf{Set})$, we refer to $(\mathbf{Set})^{\mathcal{C}^{\mathrm{op}}}$ as the **category of presheaves**, and each object is called a **presheaf**. Each presheaf can be thought of as a contravariant functor $\mathcal{C} \to (\mathbf{Set})$.

4. Let $\mathcal{C}$ be a locally small category, and fix any two objects $A, B$ in $\mathcal{C}$. We have two contravariant functors $h_A$ and $h_B$ as defined in Example 1.16.1. Now, fix any morphism $A \xrightarrow{w} B$ in $\mathcal{C}$, define the natural transformation $h_w : h_A \to h_B$ by

$$\begin{aligned} h_w(Y) : h_A(Y) = \mathrm{Hom}_{\mathcal{C}}(Y, A) &\longrightarrow h_A(Y) = \mathrm{Hom}_{\mathcal{C}}(Y, B) \\ u &\longrightarrow w \circ u \end{aligned}$$

for each object $Y$ in $\mathcal{C}$, that is, for each morphism $X \xrightarrow{f} Y$, we have the following commutative diagram:

$$\begin{tikzcd} h_A(Y) = \mathrm{Hom}_{\mathcal{C}}(Y, A) \arrow[r, "h_w(Y)"] \arrow[d, "\circ f = h_A(f)"'] & h_B(Y) = \mathrm{Hom}_{\mathcal{C}}(Y, B) \arrow[d, "\circ f = h_B(f)"] \\ h_A(X) = \mathrm{Hom}_{\mathcal{C}}(X, A) \arrow[r, "h_w(X)"'] & h_B(X) = \mathrm{Hom}_{\mathcal{C}}(X, B) \end{tikzcd}$$

We call the covariant functor $h : \mathcal{C} \to (\mathbf{Set})^{\mathcal{C}^{\mathrm{op}}}$ the **Yoneda embedding** of $\mathcal{C}$. One can see that if $h_A = h_B$ then $A = B$; that is, $h$ is "injective" in objects. The following theorem shows that $h$ is a fully faithful functor, justifying the usage "embedding".

**Theorem 1.22** (Yoneda lemma)**.** Let $\mathcal{C}$ be a locally small category and $F \in [\mathcal{C}^{\mathrm{op}}, (\mathbf{Set})]$ be a presheaf. Then for any object $X \in \mathrm{Ob}(\mathcal{C})$, we have the bijection

$$\begin{aligned} \alpha_{X,F} : \mathrm{Hom}_{[\mathcal{C}^{\mathrm{op}}, (\mathbf{Set})]}(h_X, F) &\longrightarrow F(X) \\ T &\longmapsto T(X)(\mathrm{id}_X) \end{aligned}$$

bifunctorial in $X$ and $F$. Dually, for $F \in [\mathcal{C}, (\mathbf{Set})]$ and any object $X \in \mathrm{Ob}(\mathcal{C})$, we have the bijection $\mathrm{Hom}_{[\mathcal{C}, (\mathbf{Set})]}({}_X h, F) \cong F(X)$.

*Proof.* Let $Y$ be an object of $\mathcal{C}$ and $f \in h_X(Y) = \mathrm{Hom}(X, Y)$. Put $u = T(X)(\mathrm{id}_X)$. Consider the following diagram:

$$\begin{tikzcd} f & h_X(Y) \arrow[r, "T(Y)"] & F(Y) & T(Y)(f) \arrow[dd, equal] \\ & & & F(f)(u) \\ \mathrm{id}_X \arrow[uu, mapsto] \arrow[rrr, bend right, mapsto] & h_X(X) \arrow[u] \arrow[r, "T(X)"'] & F(X) \arrow[uu, "F(f)"'] & u \arrow[u, mapsto] \end{tikzcd}$$

where the equality on the top right holds since $T$ is a natural transformation. From the diagram, we see that $T(Y)$ is completed determined by $u$ and $F$. We will make use of it to construct the inverse of $\alpha_{X,F}$; that is, given any $v \in F(X)$ and $Z \in \mathrm{Ob}(\mathcal{C})$, define

$$\begin{aligned} \beta_{X,F}(v, Z) : h_X(Z) &\longrightarrow F(Z) \\ f &\longmapsto F(f)(v) \end{aligned}$$

Then,

- $\beta_{X,F}(v, -) : h_X \to F$ is a natural transformation, and $\beta_{X,F}(u, -) = T(-)$.
- $\beta_{X,F}(v, X)(\mathrm{id}_X) = F(\mathrm{id}_X)(v) = v$.

In conclusion, $\beta_{X,F} : F(X) \to \mathrm{Hom}(h_X, F)$ is the inverse map of $\alpha_{\alpha,F}$. This completes the proof. $\square$

**Corollary 1.22.1.** Let $\mathcal{C}$ be a locally small category. Then for each pair of objects $X, Y$ of $\mathcal{C}$, there is a set-theoretic bijection

$$\begin{aligned} \mathrm{Hom}(X, Y) &\longrightarrow \mathrm{Hom}(h_X, h_Y) \\ v &\longmapsto h_v \end{aligned}$$

Hence, the Yoneda embedding of $\mathcal{C}$, defined in Example 1.21.4, is a fully faithful functor, and

$$h_X \cong h_Y \Leftrightarrow X \cong Y$$

**Definition 1.23.** Let $\mathcal{C}$ be a locally small category. A **representation** of a covariant functor $F : \mathcal{C}^{\mathrm{op}} \to (\mathbf{Set})$ is the natural isomorphism

$$\theta : h_X \to F$$

for some object $X \in \mathrm{Ob}(\mathcal{C})$.

- By Yoneda lemma, each $\theta$ is uniquely determined by an element $\xi \in F(X)$. In this case, we say the functor $F$ is **represented by a pair** $(X, \xi)$. Again, it follows from Yoneda lemma that the pair $(X, \xi)$ is determined uniquely up to unique isomorphism.
- A covariant functor $F : \mathcal{C}^{\mathrm{op}} \to (\mathbf{Set})$ is **representable** if it's represented by some pair $(X, \xi)$, where $X \in \mathrm{Ob}(\mathcal{C})$ and $\xi \in F(X)$. We say $X$ is the **represented object**, and $\xi$ is the **universal element** for $F$. In this case, by Yoneda lemma,

**Example 1.24** (Kernels)**.** Let $R$ be a ring and let $f : X \to Y$ be an $R$-module homomorphism. For any $R$-module $W$, we have the induced map

$$\begin{aligned} h_f(W) = \mathrm{Hom}_{(R\text{-}\mathbf{Mod})}(W, X) &\longrightarrow h_A(Y) = \mathrm{Hom}_{(R\text{-}\mathbf{Mod})}(W, Y) \\ u &\longrightarrow f \circ u \end{aligned}$$

Define $F(W) := \{W \xrightarrow{u} X \mid f \circ u = 0\}$, and for each homomorphism $W \xrightarrow{g} S$, let

$$\begin{aligned} F(S) &\xrightarrow{F(g)} F(W) \\ u &\longmapsto u \circ g \end{aligned}$$

Then $F : (R\text{-}\mathbf{Mod}) \to (\mathbf{Set})$ is a contravariant functor. Let $K = \ker f \subseteq X$ and $\iota : K \to X$ be the inclusion. Note that we have the "universal property" of kernel: for any $Z \xrightarrow{u} X$, there exists a unique homomorphism $Z \xrightarrow{v} K$ so that $u = \iota \circ v$, i.e, the following diagram commutes:

$$\begin{tikzcd} Z \arrow[dr, "v"] \arrow[drr, bend left, "0"] \arrow[ddr, bend right, "u"'] & & \\ & K \arrow[r, "0"] \arrow[d, "\iota"'] & Y \\ & X \arrow[ur, "f"'] & \end{tikzcd}$$

where 0 denotes the zero maps. Hence, this establishes a bijection

$$\begin{aligned} \theta(Z) : \mathrm{Hom}(Z, K) &\longrightarrow F(Z) = \{Z \xrightarrow{u} X \mid f \circ u = 0\} \\ v &\longmapsto \iota \circ v \end{aligned}$$

Clearly, $\theta(Z)$ is functorial in $Z$, which shows that $\theta : h_K \to F$ is a natural isomorphism, and that $F$ is represented by the pair $(K, \iota)$.

**Example 1.25** (Products)**.** Let $X, Y$ be sets, and consider the presheaf $F := h_X \times h_Y : (\mathbf{Set})^{\mathrm{op}} \to (\mathbf{Set})$. Then the represented object for $F$ is the Cartesian product $X \times Y$ and the universal element is the pair of projection $(\pi_X, \pi_Y)$. This reflects the "universal property" of Cartesian product: for any pair of maps $f : A \to X$ and $g : A \to Y$, there exists a unique map $h : A \to X \times Y$ so that the following diagram commutes:

$$\begin{tikzcd} & A \arrow[d, "h"] \arrow[ddl, bend right, "f"'] \arrow[ddr, bend left, "g"] & \\ & X \times Y \arrow[dl, "\pi_X"'] \arrow[dr, "\pi_Y"] & \\ X & & Y \end{tikzcd}$$

**Example 1.26.** Let For : (**Grp**) → (**Set**) be the forgetful functor. We want to show whether For is representable, that is, whether there's an natural isomorphism $\theta :_G h \to \mathrm{For}$ for some group $G$. Equivalently, for each group $H$, we need to construct a set-theoretic isomorphism

$$\theta(H) : \mathrm{Hom}_{(\mathbf{Grp})}(G, H) \longrightarrow \mathrm{For}(H)$$

This means each group homomorphism is determined uniquely by an element of $H$. One may recall that this corresponds to the universal property of free groups, and thus $(\mathbb{Z}, 1)$ is a representative of For.

## 1.4 Equivalences

**Definition 1.27.** Let $F : \mathcal{C} \to \mathcal{C}'$ be a covariant functor. $F$ is an **equivalence of categories** if there's a covariant functor $G : \mathcal{C}' \to \mathcal{C}$ such that $F \circ G \cong \mathrm{id}_{\mathcal{C}'}$ and $G \circ F \cong \mathrm{id}_{\mathcal{C}}$.

- $\mathcal{C}$ is **self-dual** if $\mathcal{C}$ and $\mathcal{C}^{\mathrm{op}}$ are equivalent.

**Proposition 1.28.** Let $F : \mathcal{C} \to \mathcal{C}'$ be a covariant functor. $F$ is an equivalence if and only if $F$ is fully faithful and **essentially surjective**, that is, for each object $Y \in \mathrm{Ob}(\mathcal{C}')$, there exists an object $X \in \mathrm{Ob}(\mathcal{C})$ such that $F(X) \cong Y$.

*Proof.* The necessity is direct. We show the sufficiency by constructing a desired functor $G : \mathcal{C}' \to \mathcal{C}$. For each $Y \in \mathrm{Ob}(\mathcal{C}')$, let $X \in \mathrm{Ob}(\mathcal{C})$ such that there's an isomorphism $\eta_Y : Y \to F(X)$ in $\mathcal{C}'$; define $G(Y) = X$.

Let $Y, Y' \in \mathrm{Ob}(\mathcal{C}')$ and $X = G(Y)$, $X' = G(Y')$. For each $f \in \mathrm{Hom}_{\mathcal{C}'}(Y, Y')$, consider the diagram

$$\begin{tikzcd}
Y \arrow[r, "\eta_Y"] \arrow[d, "f"'] & F(X) = FG(Y) \\
Y' \arrow[r, "\eta_{Y'}"] & F(X') = FG(Y')
\end{tikzcd}$$

To make it a commutative diagram, we define $FG(f) := \eta_{Y'} f \eta_Y^{-1}$ so that the diagram

$$\begin{tikzcd}
Y \arrow[r, "\eta_Y"] \arrow[d, "f"'] & F(X) = FG(Y) \arrow[d, "FG(f)"] \\
Y' \arrow[r, "\eta_{Y'}"] & F(X') = FG(X')
\end{tikzcd}$$

commutes. Since $F$ is fully faithful, we may define $G(f)$ to be the unique morphism making the above diagram commutes. Clearly, $G(\mathrm{id}_Y) = \mathrm{id}_{G(Y)}$. It suffices to show $G$ is a functor and that $GF \cong \mathrm{id}_{\mathcal{C}}$. Let $Y \xrightarrow{f} Y'$ and $Y' \xrightarrow{g} Y''$ be two morphisms in $\mathcal{C}'$. Consider the commutative diagram

$$\begin{tikzcd}
Y \arrow[r, "\eta_Y"] \arrow[d, "f"] \arrow[dd, bend right=60, "g\circ f"'] & FG(Y) \arrow[d, "FG(f)"] \arrow[dd, bend left=60, "FG(g\circ f)"] \\
Y' \arrow[r, "\eta_{Y'}"] \arrow[d, "g"] & FG(Y') \arrow[d, "FG(g)"] \\
Y'' \arrow[r, "\eta_{Y''}"] & FG(Y'')
\end{tikzcd}$$

The uniqueness shows that $G(g)G(f) = G(gf)$, and thus $G$ is a functor. It remains to show $GF \cong \mathrm{id}_{\mathcal{C}}$. Let $X \xrightarrow{f} X'$ be a morphism in $\mathcal{C}'$. Consider the commutative diagram

$$\begin{tikzcd}
F(X) \arrow[r, "\eta_{F(X)}"] \arrow[d, "F(f)"'] & FGF(X) \arrow[d, "FGF(f)"] \\
F(X') \arrow[r, "\eta_{F(X')}"'] & FGF(X')
\end{tikzcd}$$

Since $F$ is faithful, there's a unique morphism $\zeta_X \in \mathrm{Hom}_{\mathcal{C}}(X, GF(X))$ such that $F(\zeta_X) = \eta_{F(X')}$. Again, since $F$ is faithful, the above diagram implies $\zeta_{X'} \circ f = GF(f) \circ \zeta_X$, and thus $\zeta$ is a natural isomorphism from $\mathrm{id}_{\mathcal{C}}$ to $GF$. $\square$

**Example 1.29.** Let $k$ be a field. Let $\mathcal{C}$ be a category whose objects are nonnegative integers and $\mathrm{Hom}(m, n) = M_{n\times m}(k)$ for each pair $(m, n)$ of objects. Composition is defined by product of matrices. Then $\mathcal{C}$ is equivalent to the category of finite dimensional $k$-vector spaces. Indeed, we have the functor sending each $n$ to the vector space $k^n$ with standard basis, and sending each matrix to the corresponding linear map. It's clearly fully faithful, and it's essentially surjective since vector spaces of the same finite dimension are linearly isomorphic.

**Example 1.30.** Let $A, B$ be sets and $R : A \to B$ a binary relation. We may define the **converse relation** $R^\vee : B \to A$ of $R$ by

$$(b, a) \in R^\vee \Leftrightarrow (a, b) \in R$$

Note that the "converse" is really just the "reverse" of arrows. In particular, this shows that (**Rel**) and (**Rel**)$^{\mathrm{op}}$ are equivalent; precisely, the covariant functor (**Rel**) $\xrightarrow{\vee}$ (**Rel**) $\xrightarrow{\mathrm{op}}$ (**Rel**)$^{\mathrm{op}}$ is an equivalence. In conclusion, (**Rel**) is self-dual.

**Example 1.31.** Let $R$ be a ring with 1 and consider (**Mod**-$R$) and (**Mod**-$M_n(R)$), the category of right $M_n(R)$-modules. Then these two categories are equivalent, given by the functor $F$ assigning to each right

$R$-module $M$ the direct sum $M^n$ and assigning to each right $R$-module homomorphism $f$ to the diagonal homomorphism $(f, f, \ldots, f)$.

We clarify some facts:

- Let $M_n(R)$ act on $M^n$ on the right. This makes $M^n$ into a right $M_n(R)$-module.
- That $F$ is fully faithful is direct. We shall show $F$ is essentially surjective. Let $M'$ be a right $M_n(R)$-module and consider the $R$-action on $M'$ by $x'a := x'(a\delta_{ij})$ for all $x' \in M'$ and $a \in R$. Then $M := M'(\delta_{1i}\delta_{1j})$ is a $R$-submodule of $M'$, since $(a\delta_{ij}) \in Z(M_n(R))$ for all $a \in R$.

  Consider the map

$$\begin{aligned} \eta : M' &\longrightarrow M^n \\ x' &\longmapsto (x'e_{11}, x'e_{21}, \ldots, x'e_{n1}) \end{aligned}$$

  where $e_{k\ell} = (\delta_{ik}\delta_{j\ell})$ for each $k, \ell$. One can show $\eta$ is a $M_n(R)$-module isomorphism. Hence $M' \cong M^n = F(M)$.

# 2 Universal

From now on, unless stated otherwise, the term "category" always refers to "locally small category".

**Definition 2.1.** Let $\mathcal{C}$ be a category and $A$ be an object.

1. $A$ is an **initial object** if for each object $X$ in $\mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(A, X)$ is a singleton.
2. $A$ is a **terminal object**, or **final object**, if for each object $Y$ in $\mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(Y, A)$ is an singleton.
3. $A$ is a **zero object** if $A$ is both initial and final.

- An initial (resp. a terminal) object, if it exists, is unique up to a unique isomorphism.

**Example 2.2.** We list initial, terminal and zero objects in some familiar categories:

| | initial | terminal | zero |
|---|---|---|---|
| (**Set**) | empty set | singleton | - |
| (**Rel**) | empty set | empty set | empty set |
| (**Grp**) | trivial group | trivial group | trivial group |
| ($R$-**Mod**) | zero module | zero module | zero module |
| (**vect**$_k$) | zero space | zero space | zero space |
| **(Rng)** | zero ring | zero ring | zero ring |
| **(Ring)** | $\mathbb{Z}$ | zero ring | - |
| **(Field)** | - | - | - |
| **(Top)** | empty set | singleton | - |
| **(Top.)** | singleton | singleton | singleton |

**Definition 2.3.** Let $\mathcal{C}$ and $\mathcal{C}'$ be categories, $F : \mathcal{C} \to \mathcal{C}'$ be a covariant functor and $X$ be an object in $\mathcal{C}'$. An **universal arrow from** $X$ **to** $F$ is a pair $(Y, X \xrightarrow{\iota} F(Y))$ satisfying the following **universal property**: for any object $Z$ in $\mathcal{C}$ and morphism $X \xrightarrow{f} F(Z)$, there exists a unique morphism $Y \xrightarrow{g} Z$ so that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\iota"] \arrow[dr, "f"'] & F(Y) \arrow[d, "F(g)"] \\ & F(Z) \end{tikzcd}$$

Dually, an **universal arrow from** $F$ **to** $X$ can be defined in the same way, only all the arrows are reversed.

- The definition simply tells us that for each object in $\mathcal{C}'$, we have the bijection

$$\begin{aligned} {}_Y h(Z) = \mathrm{Hom}_{\mathcal{C}}(Y, Z) &\xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}'}(X, F(Z)) \\ g &\longmapsto F(g) \circ \iota \end{aligned}$$

  One can easily check that the bijection above is functorial in $Z$. Hence, we obtain a natural isomorphism ${}_Y h \to \mathrm{Hom}(X, F(-))$, and $(Y, \iota)$ is a representative of the functor $\mathrm{Hom}(X, F(-))$.

- Conversely, if $\mathrm{Hom}(X, F(-))$ is represented by $(Y, \iota)$, then for each $Z \in \mathcal{C}'$, we have the bijection

$$\begin{aligned} {}_Y h(Z) = \mathrm{Hom}_{\mathcal{C}}(Y, Z) &\xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}'}(X, F(Z)) \\ f &\longmapsto {}_X h(F(f))(\iota) = F(f) \circ \iota \end{aligned}$$

  which precisely means that $(Y, \iota)$ is the universal arrow from $X$ to $F$.

- Now suppose $(Y, F(Y) \xrightarrow{\pi} X)$ is an universal arrow from $F$ to $X$. This means for each object $Z$ in $\mathcal{C}$ we have the bijection

$$\begin{aligned} h_Y(Z) = \mathrm{Hom}_{\mathcal{C}}(Z, Y) &\xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}'}(F(Z), X) \\ g &\longmapsto \pi \circ F(g) \end{aligned}$$

  Similarly, one can show this bijection in functorial in $Z$, and thus we obtain a natural isomorphism $h_Y \to \mathrm{Hom}(F(-), X)$, and $(Y, \pi)$ is the representative of $\mathrm{Hom}(F(-), X)$; vice versa.

- If a universal arrow exists, then it's unique up to a unique isomorphism.

**Example 2.4.** Let $\mathcal{B} \xrightarrow{F} \mathcal{C} \xleftarrow{G} \mathcal{D}$ be covariant functors. We form the **comma category** $(F \downarrow G)$ as follows:

- Objects are triples $(A, B, F(A) \xrightarrow{h} G(B))$, where $A \in \mathrm{Ob}(\mathcal{B})$, $B \in \mathrm{Ob}(\mathcal{D})$.
- Morphisms from $(A, B, h)$ to $(X, Y, g)$ are pairs $(A \xrightarrow{f_1} X, B \xrightarrow{f_2} Y)$ such that the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "F(f_1)"] \arrow[d, "h"'] & F(X) \arrow[d, "g"] \\ G(B) \arrow[r, "G(f_2)"'] & G(Y) \end{tikzcd}$$

  Composition is defined in a natural way. The identity element is $(\mathrm{id}_A, \mathrm{id}_B)$ for each object $(A, B, h)$.

We now consider the situation when $\mathcal{D} = \mathbf{1}$, the category consisting of one object and one morphism. This can be viewed of as we are given $\mathcal{B} \xrightarrow{F} \mathcal{C}$ and an object $D$ of $\mathcal{C}$, and in this case, the category constructed above is denoted by $(F \downarrow D)$, has objects $(A, F(A) \xrightarrow{h} D)$, and has morphism from $(A, h)$ to $(X, g)$ of the form $A \xrightarrow{f} X$ such that the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[dr, "h"'] & F(X) \arrow[d, "g"] \\ & D \end{tikzcd}$$

Now suppose $(A, h)$ is a terminal object of $(F \downarrow D)$; this means for each object $(X, g)$, there's a unique morphism $(A, h) \xrightarrow{f} (X, g)$ such that the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[d, "h"'] & F(X) \arrow[l, "F(f)"'] \arrow[dl, "g"] \\ D & \end{tikzcd}$$

Note that this coincides with the definition of the universal arrow from $F$ to $D$.

Likewise, consider the case $\mathcal{B} = \mathbf{1}$. Suppose we are given an object $B$ in $\mathcal{C}$. Then an initial object of $(B \downarrow G)$ is the object $(A, B \xrightarrow{h} G(A))$ such that for each object $(X, B \xrightarrow{g} G(B))$, there's a unique morphism $A \xrightarrow{f} X$ so that the following diagram commutes:

$$\begin{tikzcd} G(A) \arrow[r, "G(f)"] & G(X) \\ B \arrow[u, "h"] \arrow[ur, "g"'] & \end{tikzcd}$$

Note that this coincides with the notion of the universal arrow from $B$ to $G$.

**Proposition 2.5.** Let $\mathcal{C}, \mathcal{C}'$ be categories and $F : \mathcal{C} \to \mathcal{C}'$ be a covariant functor. Let $X$ be an object of $\mathcal{C}'$. Then TFAE:

1. $(A, \iota)$ is an universal arrow from $X$ to $F$.
2. $(A, \iota)$ represents the functor $\mathrm{Hom}(X, F(-))$.
3. $(A, \iota)$ is an initial object in the comma category $(X \downarrow F)$.

Also, TFAE:

1. $(A, \pi)$ is an universal arrow from $F$ to $X$.
2. $(A, \pi)$ represents the functor $\mathrm{Hom}(F(-), X)$.
3. $(A, \pi)$ is a terminal object in the comma category $(F \downarrow X)$.

**Corollary 2.5.1.** Suppose $F : \mathcal{D} \to (\mathbf{Set})$ is a representable functor. Then the representative of $F$ is a universal arrow from $*$, the one-point set, to $F$, i.e, an initial object in the comma category $(* \downarrow F)$.

*Proof.* In view of Proposition 2.5, it suffices to note that $F \cong \mathrm{Hom}(*, F(-))$. $\square$

## 2.1 Adjoints

Let $\mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{C} \to \mathcal{D}$ be a functor. We assume for each $B \in \mathrm{Ob}(\mathcal{D})$, there's a universal arrow from $B$ to $F$; we choose one, and denote it as $(G(B), u_B)$. Then for each $A \in \mathrm{Ob}(\mathcal{C})$, we have the bijection

$$\begin{aligned} \eta_{B,A} : \mathrm{Hom}(G(B), A) &\longrightarrow \mathrm{Hom}(B, F(A)) \\ f &\longmapsto F(f) \circ u_B \end{aligned}$$

and for any fixed $B$, $\eta_{B,-} : {}_{G(B)}h(-) \to {}_{B}h(F(-))$ is a natural isomorphism. For $h \in \mathrm{Hom}_{\mathcal{D}}(B, B')$, consider the diagram

$$\begin{tikzcd} B \arrow[r, "u_B"] \arrow[d, "h"'] & F(G(B)) \\ B' \arrow[r, "u_{B'}"] & F(G(B')) \end{tikzcd}$$

By the universality, let $G(h)$ be the unique element so that $F(G(h)) \circ u_B = u_B \circ h$. As the proof of Proposition 1.28, we may show $G : \mathcal{D} \to \mathcal{C}$ is a functor, by reasons of the universality instead of being fully faithful.

We study the relations between the functors $F$ and $G$. Let $A \in \mathrm{Ob}(\mathcal{C})$. Then applying the definition of a universal arrow from $F(A)$ to $A$ to the map $g = \mathrm{id}_{F(A)}$, we obtain a unique map $\nu_A : GF(A) \to A$ such that $\mathrm{id}_{F(A)} = F(\nu_A) \circ u_{F(A)}$. Then for any $B \in \mathrm{Ob}(\mathcal{D})$, we have the map

$$\begin{aligned} \zeta_{A,B} : \mathrm{Hom}(B, F(A)) &\longrightarrow \mathrm{Hom}(G(B), A) \\ g &\longmapsto \nu_A \circ G(g) \end{aligned}$$

For each $g \in \mathrm{Hom}(B, F(A))$, we have

$$\eta_{B,A}\zeta_{A,B}(g) = F(\nu_A) \circ FG(g) \circ u_B = F(\nu_A) \circ u_{F(A)} \circ g = g$$

Since $\eta_{B,A}$ is bijective, it follows that $\zeta_{A,B} = \eta_{B,A}^{-1}$ is also bijective. This implies that $(F(A), \nu_A)$ is a universal arrow from $G$ to $A$ for all $A \in \mathrm{Ob}(\mathcal{C})$, and for any fixed $A \in \mathrm{Ob}(\mathcal{C})$, $\zeta_{A,-} : h_{F(A)}(-) \to h_A(G(-))$ is a natural isomorphism. In summary:

**Proposition 2.6.** Let $F : \mathcal{C} \to \mathcal{D}$ be a functor such that for each $B \in \mathrm{Ob}(\mathcal{D})$, there's a universal arrow from $B$ to $F$. For each $B$ choose one and call it $(G(B), u_B)$. For $h \in \mathrm{Hom}_{\mathcal{D}}(B, B')$, define $G(h) \in \mathrm{Hom}_{\mathcal{C}}(G(B), G(B'))$ to be the unique element such that $F(G(h)) \circ u_B = u_B \circ h$.

1. $G : \mathcal{D} \to \mathcal{C}$ is a functor.
2. For $A \in \mathrm{Ob}(\mathcal{C})$, there's a unique $\nu_A : GF(A) \to A$ such that $F(\nu_A)u_{F(A)} = \mathrm{id}_{F(A)}$, and $(F(A), \nu_A)$ is a universal arrow from $G$ to $A$.
3. For each $B \in \mathcal{D}$, the map $\eta_{B,-} : \mathrm{Hom}(G(B), -) \to \mathrm{Hom}(B, F(-))$ is a natural isomorphism.
4. For each $A \in \mathcal{C}$, the map $\eta_{-,A} : \mathrm{Hom}(G(-), A) \to \mathrm{Hom}(-, F(A))$ is also a natural isomorphism.

**Definition 2.7.** Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say $F$ is a **right adjoint** of $G$ and $G$ is a **left adjoint** of $F$ if for every $B \in \mathrm{Ob}(\mathcal{D})$ and $A \in \mathrm{Ob}(\mathcal{C})$, we have a bijection

$$\eta_{B,A} : \mathrm{Hom}(G(B), A) \longrightarrow \mathrm{Hom}(B, F(A))$$

and it's bifunctorial in $B$ and $A$.

- $\eta : (B, A) \mapsto \eta_{B,A}$ is called an **adjugant** from $G$ to $F$, and the triple $(F, G, \eta)$ is an **adjunction**.
- In Proposition 2.6, $G$ is simply the left adjoint of $F$.
- In Proposition 2.6, the natural transformation $u : \mathrm{id}_{\mathcal{D}} \to F \circ G$ is called the **unit** of the adjunction, and $\nu : G \circ F \to \mathrm{id}_{\mathcal{C}}$ is called the **counit**.
- Any two left adjoints $G$, $G'$ of a functor $F : \mathcal{C} \to \mathcal{D}$ are naturally isomorphic.

Conversely, we show every adjunction $(F, G, \eta)$ determines universal arrows. Fix a $B \in \mathrm{Ob}(\mathcal{D})$ and put $u_B = \eta_{B,G(B)}(\mathrm{id}_{G(B)})$. Then $\eta_{B,-} : {}_{G(B)}h \to {}_{B}h(F(-))$ is a natural isomorphism. By Yoneda lemma, for $f \in \mathrm{Hom}_{\mathcal{C}}(G(B), A)$, we have

$$\eta_{B,A}(f) = \mathrm{Hom}_{\mathcal{D}}(B, F(f))u_B = F(f) \circ u_B$$

and

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(G(B), A) &\longrightarrow \mathrm{Hom}_{\mathcal{D}}(B, F(A)) \\ f &\longmapsto F(f) \circ u_B \end{aligned}$$

is a bijection. This simply implies that $(G(B), u_B)$ is a universal arrow from $B$ to $F$.

**Example 2.8.** A **concrete category** is a category $\mathcal{C}$ together with a faithful functor $F : \mathcal{C} \to (\mathbf{Set})$. For instance, algebraic structures like $(\mathbf{Grp})$, $(\mathbf{Rng})$, etc, are concrete, with forgetful functors.

Let $(\mathcal{C}, F)$ be a concrete category and $S$ be a set. A **free object** on $S$ in $\mathcal{C}$ is an object $V \in \mathrm{Ob}(\mathcal{C})$ together with a injection $\iota : S \to F(V)$ such that for any object $Y \in \mathrm{Ob}(\mathcal{C})$ and any map $\varphi : S \to F(Y)$, there's a unique morphism $\Phi \in \mathrm{Hom}_{\mathcal{C}}(V, Y)$ so that the following diagram commutes:

$$\begin{tikzcd} F(V) \arrow[rr, "F(\Phi)"] & & F(Y) \\ & S \arrow[ul, hook, "\iota"] \arrow[ur, "\varphi"'] & \end{tikzcd}$$

This really means that $(V, S \xrightarrow{\iota} F(Y))$ is a universal arrow from $S$ to $F$. In algebra we've seen many free objects: free groups, free abelian groups, free modules, free semigroups and so on. We define the **free functor** $U : (\mathbf{Set}) \to \mathcal{C}$ to be the left adjoint of $F$, if it exists.

Consider $\mathcal{C} = (\mathbf{Top})$ and $F$ to be the forgetful functor. Then the free functor $U$ exists, and assigns to each $X$ a discrete topology. $F$ also has a right adjoint $L$, given by assigning to each $X$ a trivial topology.

**Example 2.9.** Let $R, S$ be two commutative rings with 1 and $f : R \to S$ be a homomorphism. Then every $S$-module can be seen as an $R$-module. This induces a functor $U : (S\text{-}\mathbf{Mod}) \to (R\text{-}\mathbf{Mod})$, called the restriction of scalars. One can see

$$\begin{aligned} (R - \mathbf{Mod}) &\longrightarrow (S - \mathbf{Mod}) \\ N &\longmapsto S \otimes_R N \end{aligned}$$

is the left adjoint of $U$, and

$$\begin{aligned} (R - \mathbf{Mod}) &\longrightarrow (S - \mathbf{Mod}) \\ N &\longmapsto \mathrm{Hom}_R(S, N) \end{aligned}$$

is the right adjoint of $U$, where $S$ acts on $\mathrm{Hom}_R(S, N)$ by $(sf)(s') = f(ss')$ for all $f \in \mathrm{Hom}_R(S, N)$ and $s, s' \in S$. These result from the well-known isomorphisms:

$$\mathrm{Hom}_S(S \otimes_R N, M) \cong \mathrm{Hom}_R(N, UM)$$

$$\mathrm{Hom}_S(M, \mathrm{Hom}_R(S, N)) \cong \mathrm{Hom}_R(UM, N)$$

for any fixed $R$-module $N$ and $S$-module $M$.

**Example 2.10.** Let $I$ be a set and consider the functor $- \times I : (\mathbf{Set}) \to (\mathbf{Set})$. It has a right adjoint, given by $(-)^I = {}_I h : (\mathbf{Set}) \to (\mathbf{Set})$. Indeed, for any two sets $X, Y$, we have the bijection

$$\mathrm{Hom}_{(\mathbf{Set})}(X \times I, Y) \longrightarrow \mathrm{Hom}_{(\mathbf{Set})}(X, Y^I)$$

i.e, $Y^{X \times I} \cong (Y^I)^X$. One can see the isomorphism is nature in $X$ and $Y$.

**Example 2.11.** Consider $(\mathbf{Top})$ and its full subcategory $(\mathbf{Haus})$ of Hausdorff spaces. The inclusion functor $\iota : (\mathbf{Haus}) \to (\mathbf{Top})$ has a left adjoint functor

$$H : (\mathbf{Top}) \to (\mathbf{Haus})$$

$H(X)$ is just the quotient of $X$ by the closure of the diagonal $\Delta_X \subseteq X \times X$. This follows from the fact that a Hausdorff space is precisely those having closed diagonal.

## 2.2 Products

**Definition 2.12.** Let $(X_\alpha)_{\alpha\in A}$ be an indexed family of objects in a category $\mathcal{C}$. A **product** of $(X_\alpha)_{\alpha\in A}$ is an object $P$ of $\mathcal{C}$, denoted as $\prod\limits_{\alpha\in A} X_\alpha$, together with morphisms $\left(P \xrightarrow{\pi_\alpha} X_\alpha\right)_{\alpha\in A}$, called **projections** satisfying the following property: for any object $W$ of $\mathcal{C}$ and morphisms $W \xrightarrow{f_\alpha} X_\alpha$, there exists a unique morphism $W \xrightarrow{f} P$ such that the following diagram commutes for each $\alpha \in A$:

$$\begin{tikzcd}
W \arrow[rr, "f"] \arrow[dr, "f_\alpha"'] & & P \arrow[dl, "\pi_\alpha"] \\
& X_\alpha &
\end{tikzcd}$$

- If $A$ is a finite index set, $P$, if it exists, is called a **finite product**. For instance, the product of $X, Y$, denoted as $X \times Y$ is such that for any pair of morphisms $(W \xrightarrow{f_X} X, W \xrightarrow{f_Y} Y)$, the following diagram commutes:

$$\begin{tikzcd}
& W \arrow[d, "f"] \arrow[ddl, bend right, "f_X"'] \arrow[ddr, bend left, "f_Y"] & \\
& X \times Y \arrow[dl, "\pi_X"'] \arrow[dr, "\pi_Y"] & \\
X & & Y
\end{tikzcd}$$

- A limit of $(X_\alpha)_{\alpha\in A}$, if it exists, is unique up to a unique isomorphism.

- When the corresponding products exist, we have the following isomorphisms

$$A \times B \cong B \times A$$
$$A \times (B \times C) \cong (A \times B) \times C$$

- For any category $\mathcal{C}$, we have the natural isomorphism for each object $Y$ in $\mathcal{C}$

$$\mathrm{Hom}_{\mathcal{C}}\left(\prod\nolimits_\alpha X_\alpha, Y\right) \cong \prod\nolimits_\alpha \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y)$$

**Definition 2.13.** Let $(X_\alpha)_{\alpha\in A}$ be an indexed family of objects in a category $\mathcal{C}$. A **coproduct** of $(X_\alpha)_{\alpha\in A}$ is an object $S$ in $\mathcal{C}$, denoted as $\coprod\limits_{\alpha\in A} X_\alpha$ together with morphisms $\left(X_\alpha \xrightarrow{\iota_\alpha} S\right)_{\alpha\in A}$, called **injections** satisfying the following property: given any object $W$ of $\mathcal{C}$ and morphisms $X_\alpha \xrightarrow{f_\alpha} W$, there exists a unique morphism $S \xrightarrow{f} W$ such that the following diagram commutes for each $\alpha \in A$:

$$\begin{tikzcd}
S \arrow[rr, "f"] & & W \\
& X_\alpha \arrow[ul, "\iota_\alpha"] \arrow[ur, "f_\alpha"'] &
\end{tikzcd}$$

- If $A$ is a finite index set, $P$, if it exists, is called a **finite coproduct**. For instance, the coproduct of $X, Y$, denoted as $X \sqcup Y$ is such that for any pair of morphisms $(X \xrightarrow{f_X} W, Y \xrightarrow{f_Y} W)$, the following diagram commutes:

$$\begin{tikzcd}
& W & \\
& X \sqcup Y \arrow[u, "f"] & \\
X \arrow[uur, bend left, "f_X"] \arrow[ur, "\iota_X"'] & & Y \arrow[uul, bend right, "f_Y"'] \arrow[ul, "\iota_Y"]
\end{tikzcd}$$

- A colimit of $(X_\alpha)_{\alpha \in A}$, if it exists, is unique up to a unique isomorphism.

- For any category $\mathcal{C}$, we have the natural isomorphism for each object $Y$ in $\mathcal{C}$

$$\mathrm{Hom}_{\mathcal{C}}\left(Y, \coprod_\alpha X_\alpha\right) \cong \prod_\alpha \mathrm{Hom}_{\mathcal{C}}(Y, X_\alpha)$$

**Example 2.14.** We list some familiar examples of products and coproducts:

| | product | coproduct |
|---|---|---|
| (**Set**) | Cartesian product | disjoint union |
| (**Top**) | product topology | disjoint union topology |
| (**Top.**) | product topology | wedge sum |
| (**Grp**) | direct product | free product |
| ($R$-**Mod**) | direct product | direct sum |

**Example 2.15.** We consider products and coproducts in (**Rel**), the category of relations. For any family of sets $(X_\alpha)_{\alpha \in A}$, let $\mathcal{X}$ be the disjoint union of $X_\alpha$; concretely, $\mathcal{X} = \{(x, \alpha) \mid x \in X_\alpha\}$. For each $X_\alpha$, define a relation $P_\alpha$ on $\mathcal{X} \times X_\alpha$ by

$$(x, \alpha) P_\alpha y \Leftrightarrow x = y$$

For relations $(R_\alpha : Y \to X_\alpha)_{\alpha \in A}$, we construct a relation $R : Y \to \mathcal{X}$ by

$$yR(x, \alpha) \Leftrightarrow yR_\alpha x$$

Then we have $R_\alpha = P_\alpha \circ R$ for each $\alpha$. On the other hand, any relation $S$ on $Y \times \mathcal{X}$ satisfying $R_\alpha = P_\alpha \circ S$ for each $\alpha$ is necessary that if $yR_\alpha x$, then $yS(x, \alpha)$ since $(x, \alpha)$ is the only element in $\mathcal{X}$ such that $(y, \beta)P_\alpha x$, by definition. Hence, $R$ constructed above is unique with respect to the condition of commutativity. This shows that $(\mathcal{X}, (P_\alpha)_{\alpha \in A})$ is the product of $(X_\alpha)_{\alpha \in A}$, i.e, the product in (**Rel**) is the disjoint union.

Now we consider the coproduct. For each $X_\alpha$, we define a relation $I_\alpha$ on $X_\alpha \times \mathcal{X}$ by

$$xI_\alpha(y, \beta) \Leftrightarrow x = y \wedge \alpha = \beta$$

For relations $(R_\alpha : X_\alpha \to Y)_{\alpha \in A}$, we construct a relation $R : \mathcal{X} \to Y$ by

$$(x, \alpha)Ry \Leftrightarrow xR_\alpha y$$

Then we have $R_\alpha = R \circ I_\alpha$. Likewise, any relation $S : \mathcal{X} \to Y$ satisfying $R_\alpha = S \circ I_\alpha$ is necessary that if $xR_\alpha y$, then $yS(x, \alpha)$ since $(x, \alpha)$ is the only element in $\mathcal{X}$ such that $xI_\alpha(y, \beta)$; $R$ is unique. This shows $(\mathcal{X}, (I_\alpha)_{\alpha \in A})$ is the coproduct of $(X_\alpha)_{\alpha \in A}$.

Hence, in the category of relations, products and coproducts coincides. In fact, it's just a corollary of the self-dualness of (**Rel**). (see Example 1.30)

**Example 2.16.** Let $X$ be a set and $\leqslant$ be a partial order on $X$. Recall that a partial order on $X$ satisfies, for $x, y, z \in X$

- $x \leqslant x$
- $x \leqslant y$ and $y \leqslant z$ imply $x \leqslant z$
- $x \leqslant y$ and $y \leqslant x$ imply $x = y$

Such a pair $(X, \leqslant)$ is called a **poset**. Equivalently, we may think of a poset as a category such that

- for each pair of objects $x, y$, there is at most one morphism from $x$ to $y$
- for each pair of objects $x, y$, if there are a morphism from $x$ to $y$ and a morphism from $y$ to $x$, then $x = y$.

that is, elements of $X$ are as objects, and the order relations are as morphisms.

Let $S$ be a subset of $X$. A **meet** of $S$ is an element $s \in S$, denoted as $\bigwedge S$, such that for $w, x, y \in S$

1. $s \leqslant x$
2. if $w \leqslant x$ and $w \leqslant y$, then $w \leqslant s$.

In other words, $s$ is the infimum of $S$. A **join** of $S$ is the supremum of $S$, denoted as $\bigvee S$. Note that both join and meet of $S$ may not exist. Just as their definitions show, $\bigwedge S$ is the product of elements of $S$, and $\bigvee S$ is the coproduct. In particular, a poset $X$ has an initial (resp. a terminal) object if and only if a meet (resp. a join) of $X$ exist.

## 2.3 Equalizers

**Definition 2.17.** Let $\mathcal{C}$ be a category and $A \underset{g}{\overset{f}{\rightrightarrows}} B$ be a parallel of morphisms in $\mathcal{C}$. An **equalizer of** $f, g$ is an object $K$ in $\mathcal{C}$ with a morphism $K \xrightarrow{k} A$ such that $f \circ k = g \circ k$ and satisfying the following property: for any pair $(M, M \xrightarrow{m} A)$ such that $f \circ m = g \circ m$, there exists a unique morphism $M \xrightarrow{n} K$ such that $m = k \circ n$, i.e, the following diagram commutes:

$$\begin{tikzcd}
M \arrow[d, "n"'] \arrow[dr, "m"] & & \\
K \arrow[r, "k"'] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B
\end{tikzcd}$$

Dually, one can define a **coequalizer** of $f, g$ by reversing all arrows involved.

- If an equalizer (resp. a coequalizer) exists, it is unique up to a unique isomorphism. By $\ker(f, g)$ we denote their equalizer, and by $\operatorname{coker}(f, g)$ we denote their coequalizer.
- We have the natural isomorphism

$$\operatorname{Hom}_{\mathcal{C}}(M, \ker(f, g)) \cong \{u \in \operatorname{Hom}_{\mathcal{C}}(M, A) \mid f \circ u = g \circ u\}$$

$$\operatorname{Hom}_{\mathcal{C}}(\operatorname{coker}(f, g), M) \cong \{u \in \operatorname{Hom}_{\mathcal{C}}(Y, M) \mid u \circ f = u \circ g\}$$

**Property 2.18.** Suppose $A \underset{g}{\overset{f}{\rightrightarrows}} B$ have an equalizer $(K, k)$. Then $k : K \to A$ is a monomorphism. Dually, a coequalizer is an epimorphism.

**Property 2.19.** An arrow $A \xrightarrow{f} B$ that is both an epimorphism and an equalizer is an isomorphism.

**Example 2.20.** In (**Set**), (**Top**), (**Grp**), and so on, the equalizer of two morphisms $A \underset{g}{\overset{f}{\rightrightarrows}} B$ is just given by

$$\ker(f, g) = \{a \in A \mid f(a) = g(a)\}$$

provided with the structure induced by that of $A$.

**Example 2.21.** In (**Set**), the coequalizer of $A \underset{g}{\overset{f}{\rightrightarrows}} B$ is the quotient of $B$ by the equivalence relation generated by the pairs $(f(a), g(a))$ for each $a \in A$. In (**Top**), it is equipped with the quotient topology.

**Example 2.22.** In (**Grp**), (**Ring**), the coequalizer of $A \underset{g}{\overset{f}{\rightrightarrows}} B$ is the quotient of $B$ by the normal subgroup (resp. ideal) generated by all the pairs $(f(a), g(a))$ for $a \in A$. In (**AbGrp**), since each subgroup is normal, the coequalizer is just the quotient of $B$ by $(f - g)(A)$. This is also true in ($R$-**Mod**).

**Example 2.23.** In (**Haus**), the category of Hausdorff spaces, with continuous maps as morphisms, the coequalizer of two morphisms $A \underset{g}{\overset{f}{\rightrightarrows}} B$ is the quotient of $B$ by the closure $\overline{R} \subseteq B \times B$ of the equivalence relation $R$ generated by all the pairs $(f(a), g(a))$ for $a \in A$. Indeed, a quotient of of a Hausdorff space by a closed equivalence relation is still a Hausdorff space. Choose $C$ a Hausdorff space and $h : B \to C$ as continuous map such that $h \circ f = h \circ g$. The diagonal $\Delta_C \subseteq C \times C$ is closed, and thus $(h \times h)^{-1}(\Delta_C)$ is a closed equivalence relation on $B$ containing all the pairs $(f(a), g(a))$ for $a \in A$; thus it contains $\overline{R}$, and $h$ factors through the quotient. This remains true in the category of compact Hausdorff spaces.

## 2.4 Pullbacks

**Definition 2.24.** Let $A \xrightarrow{f} C \xleftarrow{g} B$ be morphisms in a category $\mathcal{C}$. A **pullback** of $(f, g)$ is a triple $(P, A \xleftarrow{g'} P \xleftarrow{f'} B)$ with $f \circ g' = g \circ f'$ satisfying the following property: for any triple $(Q, A \xleftarrow{g''} Q \xleftarrow{f''} B)$ with $f \circ g'' = g \circ f''$, there exists a unique morphism $q : Q \to P$ making the following diagram commute:

$$\begin{tikzcd}
Q \arrow[dr, "q"] \arrow[drr, bend left, "f''"] \arrow[ddr, bend right, "g''"'] & & \\
& P \arrow[r, "f'"] \arrow[d, "g''"'] & B \arrow[d, "g"] \\
& A \arrow[r, "f"'] & C
\end{tikzcd}$$

Dually, one can define a **pushout** of $A \xleftarrow{f} C \xrightarrow{g} B$ by reversing all the arrows involved.

- If a pullback (resp. a pushout) exists, it is unique up to a unique isomorphism. Also, denote by $A \times_C B$ the pullback of $f, g$.

**Property 2.25.** Let $(P, f', g')$ be a pullback of $(f, g)$.

1. If $g$ is a monomorphism, so is $g'$.
2. If $g$ is an isomorphism, so is $g'$.

That is, the pullback of a monomorphism is a monomorphism. Dually, the pushout of an epimorphism is an epimorphism.

**Example 2.26.** In general, the pullback of an epimorphism is not an epimorphism. Consider an epimorphism $f : X \to Y$ in the category of Hausdorff spaces. In Example 1.9, we know $f(X)$ is dense in $Y$; suppose $f$ is not surjective. Pick $y \in Y \backslash f(X)$. Then the pullback of $y$ and $f(X)$ is just the empty space.

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \{*\} \arrow[r, "y"'] & Y \end{tikzcd}$$

However, the empty space is not dense in the singleton, and thus it's not an epimorphism.

**Example 2.27.** Let $F, G, H$ be groups and $G \xleftarrow{g} F \xrightarrow{h} H$ be injective homomorphisms. The pushout of them is the free product with amalgamation

$$(G * H)/N$$

where $N$ is the normal subgroup generated by all the element of the form $g(x)h(x)^{-1}$ for $x \in F$.

**Example 2.28.** Let $(X, x_0)$ be a connected point space that is also a union of path-connected open sets $A, B$, each containing $x_0$ and with $A \cap B$ path-connected. Then the pushout of $\pi_1(A) \xleftarrow{\iota_A} \pi_1(A \cap B) \xrightarrow{\iota_B} \pi_1(B)$ is the triple $(\pi_1(X), \pi_1(A) \xrightarrow{j_A} \pi_1(X) \xleftarrow{j_B} \pi_1 B)$.

$$\begin{tikzcd} \pi_1(A \cap B) \arrow[r, "\iota_A"] \arrow[d, "\iota_B"'] & \pi_1(A) \arrow[d, "j_A"] \\ \pi_1(B) \arrow[r, "j_B"'] & \pi_1(X) \end{tikzcd}$$

This is the van Kampen Theorem for fundamental groups in algebraic topology. In fact, we have the isomorphism

$$\pi_1(X) \cong (\pi_1(A) * \pi_1(B))/N$$

where $N$ is the normal subgroup generated by all elements of the form $\iota_A(x)\iota_B(x)^{-1}$ for $x \in \pi_1(A \cap B)$.

**Example 2.29.** Let $X, Y$ be topological spaces, $A \subseteq Y$ a closed subspace, and $f : A \to X$ a continuous map. The pushout of $f : A \to X$ and the inclusion $A \subseteq Y$ is the adjunction space

$$X \cup_f Y = (X \sqcup Y)/\sim$$

where $\sim$ is the equivalence relation generated by $a \sim f(a)$ for $a \in A$.

**Example 2.30.** Let $A, B, C$ be sets, $f : A \to C$ and $g : B \to C$ be maps. Then the pullback of $f, g$ is

$$A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\} = \bigcup_{c \in f(A) \cap g(B)} f^{-1}(c) \times g^{-1}(c)$$

This is also called the **fibre product** of $f, g$, named after the last equality.

**Example 2.31.** Let $A, B$ be sets, $f : A \to B$, and $B_0 \subseteq B$. Then the pullback of $f$ and the inclusion $B_0 \subseteq B$ is the inverse image $f^{-1}(B_0)$, together with the inclusion $f^{-1}(B_0) \subseteq A$ and the restriction of $f$ to $f^{-1}(B_0)$.

**Definition 2.32.** In a category $\mathcal{C}$, the **kernel pair** of a morphism $A \xrightarrow{f} B$ is the pullback $(P, \alpha, \beta)$ of $f$ with itself.

- If the kernel pair of $f$ exists, $\alpha$ and $\beta$ are epimorphisms.

**Example 2.33.** Let $A, B$ be sets and $f : A \to B$ be a map. The kernel pair $(P, \alpha, \beta)$ of $f$ is given by

$$P = A \times_B A = \{(a_1, a_2) \in A \times A \mid f(a_1) = f(a_2)\}$$

$$\alpha(a_1, a_2) = a_1,\ \beta(a_1, a_2) = a_2$$

Note that $P$ is the equivalence relation on $A$ determined by $f$. In other words, the image $f(A)$ of $f$ is the quotient of $A$ by $P$.

**Proposition 2.34.** Let $A \xrightarrow{f} B$ be a morphism. TFAE:

1. $f$ is a monomorphism.
2. the kernel pair of $f$ exists and is given by $(A, \mathrm{id}_A, \mathrm{id}_A)$
3. the kernel pair $(P, \alpha, \beta)$ of $f$ exists and is such that $\alpha = \beta$

**Proposition 2.35.** If a coequalizer has a kernel pair, it's the coequalizer of its kernel pair.

**Proposition 2.36.** If a kernel pair has a coequalizer, it's the kernel pair of its coequalizer.

**Proposition 2.37.** Let $\mathcal{C}$ be a category, and consider the commutative diagram:

$$\begin{tikzcd}
A \arrow[r, "a"] \arrow[d, "c"'] & B \arrow[r, "b"] \arrow[d, "d"] & C \arrow[d, "e"] \\
D \arrow[r, "f"'] & E \arrow[r, "g"'] & F
\end{tikzcd}$$

Suppose the left rectangle is a pullback. Then the outer rectangle is a pullback if and only if the left rectangle is a pullback.

# 3 Limits

**Definition 3.1.** Let $F : J \to \mathcal{C}$ be a covariant functor. The **limit** of $F$ is an object $L$, denoted by $\varprojlim F$ of $\mathcal{C}$ together with a family of morphisms $\varphi_X : L \to F(X)\,(X \in \mathrm{Ob}(J))$ satisfying the following properties:

1. If $X \xrightarrow{\alpha} Y$ is a morphism in $J$, then $\varphi_Y = F(\alpha) \circ \varphi_X$, i.e, the following diagram commutes:

$$\begin{tikzcd} & L \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\ F(X) \arrow[rr, "F(\alpha)"'] & & F(Y) \end{tikzcd}$$

2. If $N$ is another object, together with morphisms $\psi_X : M \to F(X)\,(X \in \mathrm{Ob}(J))$ satisfying property 1., then there exists a unique morphism $N \xrightarrow{u} L$ such that $\psi_X = \varphi_X \circ u$ for each object $X$ of $J$, i.e, the following diagram commutes:

$$\begin{tikzcd} & N \arrow[d, "u"] \arrow[ddl, bend right, "\psi_X"'] \arrow[ddr, bend left, "\psi_Y"] & \\ & L \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\ F(X) \arrow[rr, "F(\alpha)"'] & & F(Y) \end{tikzcd}$$

   In other words, $L$ is *final* with respect to property 1.

Dually, we define the **colimit** of $F$, denoted by $\varinjlim F$, to be the limit of the functor $J^{\mathrm{op}} \xrightarrow{\mathrm{op}} J \xrightarrow{F} C \xrightarrow{\mathrm{op}} C^{\mathrm{op}}$. That is, the colimit of $F$ is defined in the way that all the arrows involved are reversed.

- One should think the category $J$ as the category of *indices*, a generalization of index sets. Then a functor $F : J \to \mathcal{C}$ is nothing but a numbering of $\mathcal{C}$ by $J$.

- For all $N \in \mathrm{Ob}(\mathcal{C})$, we have the bijection

$$\mathrm{Hom}_{\mathcal{C}}(N, \varprojlim F) \cong \varprojlim \mathrm{Hom}_{\mathcal{C}}(N, F(-))$$

**Example 3.2** (cones and cocones)**.** Any object $N$ together with morphisms $\psi_X : M \to F(X)\,(X \in \mathrm{Ob}(J))$ satisfying property 1. is called a **cone** from $N$ to $F$. By $\mathrm{Cone}(N, F)$ we mean the collection of cones from $N$ to $F$.

We may construct a **category of cones to** $F$ as a comma category: Let $J, \mathcal{C}$ be categories. Define the **diagonal functor** $\Delta : \mathcal{C} \to \mathcal{C}^J$ by sending each object $A$ to the **constant diagram** $\Delta_A : J \to \mathcal{C}$ to $A$, and each morphism $A \xrightarrow{f} B$ to the natural transformation $\Delta_A \to \Delta_B$ (by assigning each morphism to $f$). Then the comma category $(\Delta \downarrow F)$ is precisely the category of cones to $F$:

- objects are pairs $(A, \Delta_A \xrightarrow{\varphi} F)$, where the $A$ are objects in $\mathcal{C}$. This means for each morphism $X \xrightarrow{\alpha} Y$ in $J$, we have the following commutative diagram:

$$\begin{tikzcd} & A \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\ F(X) \arrow[rr, "F(\alpha)"'] & & F(Y) \end{tikzcd}$$

  In other words, an object of $(\Delta \downarrow F)$ is a cone to $F$.

- morphisms from $(A, \psi)$ to $(B, \varphi)$ are $A \xrightarrow{f} B$ such that for each morphism $X \xrightarrow{\alpha} Y$, the following diagram commutes:

$$\begin{tikzcd} & A \arrow[d, "f"] \arrow[ddl, bend right, "\psi_X"'] \arrow[ddr, bend left, "\psi_Y"] & \\ & B \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\ F(X) \arrow[rr, "F(\alpha)"'] & & F(Y) \end{tikzcd}$$

Then, a limit of $F$ is precisely a terminal object in the comma category $(\Delta \downarrow F)$, i.e, an universal arrow from $\Delta$ to $F$ (thought of as an object in $C^J$).

Similarly, we have the category of **cocones** $(F \downarrow \Delta)$:

- objects are pairs $(A, F \xrightarrow{\varphi} \Delta_A)$, where the $A$ are objects in $\mathcal{C}$. This means for each morphism $X \xrightarrow{\alpha} Y$ in $J$, we have the following commutative diagram:

$$\begin{tikzcd} & A & \\ F(X) \arrow[ur, "\varphi_X"] \arrow[rr, "F(\alpha)"'] & & F(Y) \arrow[ul, "\varphi_Y"'] \end{tikzcd}$$

- morphisms from $(A, \psi)$ to $(B, \varphi)$ are $A \xrightarrow{f} B$ such that for each morphism $X \xrightarrow{\alpha} Y$, the following diagram commutes:

$$\begin{tikzcd}
 & B & \\
 & A \arrow[u, "f"] & \\
F(X) \arrow[uur, bend left, "\varphi_X"] \arrow[ur, "\psi_X"] \arrow[rr, "F(\alpha)"'] & & F(Y) \arrow[uul, bend right, "\varphi_Y"'] \arrow[ul, "\psi_Y"']
\end{tikzcd}$$

Then, a colimit of $F$ is precisely an initial object in the comma category $(F \downarrow \Delta)$, i.e, an universal arrow from $F$ to $\Delta$.

**Example 3.3.** Let $J$ be a category and let $\mathcal{C}$ be a category such that every diagram of shape $J$ has a limit. In this case, we may define a **limit functor**

$$\lim : \mathcal{C}^J \to \mathcal{C}$$

assigning to each diagram to its limit and assigning each natural transformation $\eta : F \to G$ to the unique morphism $\lim \eta : \lim F \to \lim G$ so that for each $X \xrightarrow{\alpha} Y$ the following diagram commutes

$$\begin{tikzcd}
G(X) \arrow[rr, "G(\alpha)"] & & G(Y) \\
 & \lim G \arrow[ul, "\psi_X"'] \arrow[ur, "\psi_Y"] & \\
 & \lim F \arrow[u, "\lim \eta"] \arrow[dl, "\varphi_X"'] \arrow[dr, "\varphi_Y"] & \\
F(X) \arrow[uuu, "\eta(X)"] \arrow[rr, "F(\alpha)"] & & F(Y) \arrow[uuu, "\eta(Y)"']
\end{tikzcd}$$

In view of Proposition 2.5, 2.6 and Example 3.2, we see that lim is the right adjoint functor of the diagonal functor $\Delta$, and for each $N \in \mathrm{Ob}(\mathcal{C})$, we have the bijection

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{C}}(N, \lim F) &\longrightarrow \mathrm{Hom}_{\mathcal{C}^J}(\Delta_N, F) = \mathrm{Cone}(N, F) \\
g &\longmapsto \varphi \circ \Delta(g)
\end{aligned}$$

bifunctorial in $N$ and $F$.

Dually, let $\mathcal{C}$ be a category such that every diagram of shape $J$ has a colimit. Similarly, we may define a **colimit functor** $\mathrm{colim} : \mathcal{C}^j \to \mathcal{C}$, and we see colim is the left adjoint functor of the diagonal functor $\Delta$, and for each $N \in \mathrm{Ob}(\mathcal{C})$, we have the bijection

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(\mathrm{colim} F, N) &\longrightarrow \mathrm{Hom}_{\mathcal{C}^J}(F, \Delta_N) = \mathrm{Cocone}(N, F) \\ g &\longmapsto \Delta(g) \circ \varphi \end{aligned}$$

bifunctorial in $N$ and $F$.

**Example 3.4** (Terminal objects)**.** Let $\mathcal{C}$ be a category and $J$ be the empty category, that is, the category consisting of no object and no morphism. Let $F : J \to \mathcal{C}$ be a diagram. Then the limit $\varprojlim F$ is just the terminal object in $\mathcal{C}$.

**Example 3.5** (Products)**.** Let $\mathcal{C}$ be a category and $J$ be the category consisting of two objects $\mathbf{1}, \mathbf{2}$ with only identity morphisms. (A category with only identity morphisms is called a **discrete category**.) Let $F : J \to \mathcal{C}$ be a diagram and put $A_1 = F(\mathbf{1})$, $A_2 = F(\mathbf{2})$. Then the limit $\varprojlim F$ is just the product of $A_1$ and $A_2$ in $\mathcal{C}$. In general, let $J$ be a discrete small category and $F : J \to \mathcal{C}$ be a diagram. Put $A_j = F(j)\,(j \in \mathrm{Ob}(J))$. Then the limit $\varprojlim F$ is just the product of $(A_j)_{j \in \mathrm{Ob}(J)}$.

**Example 3.6** (Equalizers)**.** Let $J$ be a category with two objects $\mathbf{1}, \mathbf{2}$, with morphisms like

$$\begin{tikzcd} \mathbf{1} \arrow[loop left] \arrow[r, bend left, "\alpha"] \arrow[r, bend right, "\beta"'] & \mathbf{2} \arrow[loop right] \end{tikzcd}$$

Then for any diagram $F : J \to \mathcal{C}$, $\varprojlim F$ is the equalizer of morphisms $F(\alpha), F(\beta)$. In general, let $I$ be a set and $J$ be a category with two objects $\mathbf{1}, \mathbf{2}$ such that

$$\mathrm{Hom}_J(\mathbf{1}, \mathbf{2}) = \{f_i\}_{i \in I},\ \mathrm{Hom}_J(\mathbf{2}, \mathbf{1}) = \varnothing,\ \mathrm{Hom}_J(\mathbf{1}, \mathbf{1}) = \{\mathrm{id}\},\ \mathrm{Hom}_J(\mathbf{2}, \mathbf{2}) = \{\mathrm{id}\}.$$

Then for any diagram $F : J \to \mathcal{C}$, the limit $\varprojlim F$ is the equalizer of the family of morphisms $\{F(f_i)\}_{i \in I}$.

**Example 3.7** (Pullbacks)**.** Let $J$ be a category with three objects $\mathbf{1}, \mathbf{2}, \mathbf{3}$, with morphisms like

$$\begin{tikzcd} \mathbf{1} \arrow[loop left] \arrow[r, "\beta"] & \mathbf{2} \arrow[loop above] & \mathbf{3} \arrow[loop right] \arrow[l, "\alpha"'] \end{tikzcd}$$

Then for any diagram $F : J \to \mathcal{C}$, $\varprojlim F$ is the pullback of morphisms $F(\alpha), F(\beta)$.

**Example 3.8** (Inverse limits)**.** Let $J$ be a small category induced by a poset $(X, \leqslant)$ in which every two elements has an upper bound. (see Example 2.16.) Let $F : J^{\mathrm{op}} \to \mathcal{C}$ be a diagram, and put $A_i = F(i)$ for each $i \in X$. The limit $\varprojlim F$, if it exists, is called the **inverse limit** of objects $\{A_i\}_{i \in X}$, also denoted as $\varprojlim_i A_i$, when the morphisms are clear from the context.

In (**Set**), the inverse limit exists: Let $\{A_i\}_{i \in I}$ be sets indexed by a poset such that for each for each $i \leqslant j$, there's a map $\mu_{ji} : A_j \to A_i$ such that

1. $\mu_{ji} \circ \mu_{kj} = \mu_{ki}$ for $i \leqslant j \leqslant k$,

2. $\mu_{ii} = 1$ for all $i \in I$.

We say it's an **inverse system** on the family $\{A_i\}_{i \in I}$. Then the inverse limit $\varprojlim_i A_i$ is given by

$$\left\{ (a_i)_{i \in I} \in \prod_i A_i \mid \mu_{ji}(a_j) = a_i \text{ for } i \leqslant j \right\}$$

together with projections $\pi_j : \varprojlim_i A_i \to A_j$ for each $j \in I$.

In (**AbGrp**), (**Rng**) and ($R$-**Mod**), the inverse limits exist, and are constructed in the same way as (**Set**). In (**Top**), the inverse limit exists, and is given by the one in (**Set**) equipped with the initial topology induced by $\{\pi_j\}_{j \in I}$.

Sometimes an inverse limit, though exists, may be empty. There are some sufficient condition to guarantee the nonemptyness. For instance, if $\{A_i\}_{i \in I}$ is an inverse system of finite sets, then $\varprojlim_i A_i$ is not empty. More generally, if $\{A_i\}_{i \in I}$ is an inverse system in the category of compact Hausdorff spaces, then $\varprojlim_i A_i$ is not empty.

**Lemma 3.9.** Let $(A_i, \varphi_{ji})$ is an inverse system of Hausdorff spaces and for each $j \in I$, put

$$Y_j = \left\{ (x_i)_{i \in I} \in \prod_i A_i \mid \varphi_{jk}(x_j) = x_k \text{ for all } k \leqslant j \right\}$$

Then $Y_j$ is a nonempty closed subset of $\prod_i A_i$.

*Proof.* That $Y_j$ is nonempty is clear. Pick $(x_i)_{i \in I} \in \prod_i A_i \backslash Y_j$. Then there exists $i \leqslant j$ such that $\varphi_{ji}(x_j) \neq x_i$. Let $U', U$ be disjoint open neighborhoods of $\varphi_{ij}(x_j)$ and $x_i$ respectively, and let $V$ be an open neighborhood of $x_j$ such that $\varphi_{ij}(V) \subseteq U'$. Consider the basis element $\prod_k U_k$, with $U_i = U$, $U_j = V$ and $U_k = A_k$, $k \neq i, j$; it's an open neighborhood of $(x_i)_{i \in I}$ disjoint from $Y_j$. □

**Proposition 3.10.** If $(A_i, \varphi_{ij})$ is an inverse system of compact Hausdorff spaces, then $\varprojlim_i A_i$ is nonempty.

*Proof.* For each $j \in I$, put $Y_j = \{(x_i)_{i \in I} \in \prod_i A_i \mid \varphi_{jk}(x_j) = x_k \text{ for all } k \leqslant j\}$. By the lemma above, the $Y_j$ are nonempty and closed. Note that $Y_i \subseteq Y_j$ whenever $i \geqslant j$. Since $I$ is directed, it follows that $\{Y_i\}_{i \in I}$ has the finite intersection property. By Tychonoff's theorem, $\prod_i X_i$ is compact, and so are the $Y_i$. Since

$$\varprojlim_i A_i = \bigcap_{j \in I} Y_j$$

the result follows. □

## 3.1 Existence theorem for limits

**Definition 3.11.** A category is **complete** if every small diagram has a limit. A category is **finitely complete** if every finite diagram has a limit.

- By a **small limit** we meas a limit of some small diagram. Similar for a **finite limit**.

Consider a functor $F : \mathcal{D} \to (\mathbf{Set})$, where $\mathcal{D}$ is small. One can easily see the set

$$L = \left\{ (x_D)_{D \in \mathcal{D}} \in \prod_{D \in \mathcal{D}} F(D) \,\middle|\, F(f)(x_D) = x_{D'} \text{ for all } f : D \to D' \right\}$$

together with the obvious projections $p_D : L \to F(D)$ is the limit of $F$. Let

$$\prod_{D \in \mathcal{D}} F(D) \underset{\beta}{\overset{\alpha}{\rightrightarrows}} \prod_{f \in \mathcal{D}} F(c(f))$$

defined by

$$\begin{aligned} \alpha((x_D)_{D \in \mathcal{D}}) &= (x_{c(f)})_{f \in \mathcal{D}} \\ \beta((x_D)_{D \in \mathcal{D}}) &= (F(f)(x_{d(f)}))_{f \in \mathcal{D}} \end{aligned}$$

where $d(f)$, $c(f)$ mean the domain and the codomain of $f$, respectively. Note that $L$ is the equalizer of $\alpha, \beta$:

$$L \hookrightarrow \prod_{D \in \mathcal{D}} F(D) \underset{\beta}{\overset{\alpha}{\rightrightarrows}} \prod_{f \in \mathcal{D}} F(c(f))$$

We will see that this construction generalizes to an arbitrary category, from which the existence of limits will follow from that of products and equalizer.

**Theorem 3.12.** Let $\mathcal{C}$ be a category. TFAE:

1. $\mathcal{C}$ is complete.
2. $\mathcal{C}$ admits all small products and all binary equalizers.

*Proof.* That 1. implies 2. is already demonstrated in Example 3.5 and 3.6. Conversely, let $\mathcal{D}$ be a small category and $F : \mathcal{D} \to \mathcal{C}$ a functor. We construct the products

$$\left( P_d = \prod_{D \in \mathcal{D}} F(D), (p'_D)_{D \in \mathcal{D}} \right) \text{ and } \left( P_c = \prod_{f \in \mathcal{D}} (F(c(f))), (p''_f)_{f \in \mathcal{D}} \right)$$

put $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(P_d, P_c)$ to be the unique morphisms so that $p''_f \circ \alpha = p'_{c(f)}$ and $p''_f \circ \beta = F(f) \circ p'_{d(f)}$ for each $f$ and let $(L, \ell)$ be the equalizer of the pair $(\alpha, \beta)$.

$$
\begin{tikzcd}
M \arrow[r, "q_D"] \arrow[d, shift right, "q"'] \arrow[d, shift left, "\bar{q}"] \arrow[dr, "q'"] & F(D) & F(c(f)) \\
L \arrow[r, "\ell"] \arrow[dd, "p_D"] & P_d \arrow[u, "P'_D"] \arrow[ur, "p'_{c(f)}"] \arrow[r, shift left, "\alpha"] \arrow[r, shift right, "\beta"'] \arrow[dl, "p'_D"] \arrow[dd, "F(c(f))"] & P_c \arrow[u, "p''_f"'] \arrow[dd, "p''_f"] \\
F(D) & & \\
 & F(d(f)) \arrow[r, "F(f)"'] & F(c(f))
\end{tikzcd}
$$

Define $p_D = p'_D \circ \ell$. We contend that $(L, (p_D)_{D \in \mathcal{D}})$ is the limit of the functor $F$.

- $(L, (p_D)_{D \in \mathcal{D}})$ is a cone on $F$. Indeed, for any morphism $D \xrightarrow{f} D'$ in $\mathcal{D}$, we have

$$F(f) \circ p_D = F(f) \circ p'_D \circ \ell = p''_f \circ \beta \circ \ell = p''_f \circ \alpha \circ \ell = p'_{D'} \circ \ell = p_{D'}$$

- If $(M, (q_D)_{D \in \mathcal{D}})$ is another cone on $F$, there's a unique $q' \in \mathrm{Hom}_{\mathcal{C}}(M, P_d)$ so that $q_D = p'_D \circ a'$ for all $D \in \mathcal{D}$. But for every $D \xrightarrow{f} D'$ in $\mathcal{D}$, we have

$$\begin{aligned}
p''_f \circ \alpha \circ q' &= p'_{D'} \circ q' \\
&= q_{D'} \\
&= F(f) \circ q_D \\
&= F(f) \circ p'_D \circ q' = p''_f \circ \beta \circ q'
\end{aligned}$$

  from which $\alpha \circ q' = \beta \circ q'$, implying there's a unique morphism $q \in \mathrm{Hom}_{\mathcal{C}}(M, L)$ such that $\ell \circ q = q'$. Thus

$$p_D \circ q = p'_D \circ \ell \circ q = p'_D \circ q' = q_D$$

  as wanted.

- It remains to show the uniqueness of $q$. Let $\bar{q}$ be another morphism such that $p_D \circ \bar{q} = q_D$ for all $D \in \mathcal{D}$. Since $\ell$ is a monomorphism, it suffices to show $\ell \circ q = \ell \circ q'$, and it's equivalent to proving $p'_D \circ \ell \circ q = p'_D \circ \ell \circ q'$, which holds since

$$p'_D \circ \ell \circ q = p'_D \circ q' = q_D = p_D \circ \bar{q} = p'_D \circ \ell \circ \bar{q}$$

$\square$

**Proposition 3.13.** Let $\mathcal{C}$ be a category. TFAE:

1. $\mathcal{C}$ is finitely complete.

2. $\mathcal{C}$ has a terminal object, binary products and equalizers.

3. $\mathcal{C}$ has a terminal object and pullbacks.

*Proof.* In view of Example 3.4, that 1. $\Rightarrow$ 2., 3. is clear. Now assume 2. By associativity of products, the product of a finite nonempty family of objects exists as long as binary products do. Observing the proof of Theorem 3.12, notice that when $\mathcal{D}$ is finite, so are the products involved in the proof.

Now we assume 3. Note that the product of $A, B$ is the pullback of $A \to 1 \leftarrow B$

$$\begin{tikzcd} A \times B \arrow[r] \arrow[d] & B \arrow[d] \\ A \arrow[r] & 1 \end{tikzcd}$$

Let $A \underset{g}{\overset{f}{\rightrightarrows}} B$ be a pair of parallel arrows. Consider the pullback

$$\begin{tikzcd} P \arrow[r, "\ell"] \arrow[d, "k"'] & A \arrow[d, "\mathrm{id}_A \times g"] \\ A \arrow[r, "\mathrm{id}_A \times f"'] & A \times B \end{tikzcd}$$

We have

$$k = p_A \circ (\mathrm{id}_A \times f) \circ k = p_A \circ (\mathrm{id}_A \times g) \circ \ell = \ell$$

and

$$f \circ k = p_B \circ (\mathrm{id}_A \times g) \circ k = p_B \circ (\mathrm{id}_A \times g) \circ \ell = g \circ \ell = g \circ k$$

Moreover, if $X \xrightarrow{\psi} A$ is such that $f \circ \psi = g \circ \psi$, one has $(\mathrm{id}_A \times f) \circ \psi = (\mathrm{id}_A \times g) \circ \psi$, so there's a unique morphism $\varphi \in \mathrm{Hom}(X, P)$ such that $\psi = k \circ \varphi = \ell \circ \varphi$. This shows $k = \ker(f, g)$. $\square$

**Definition 3.14.** Let $F : \mathcal{C} \to \mathcal{C}'$ be a functor, $G \in \mathcal{C}^J$ a diagram and $L$ a cone to $G$.

1. $F$ **preserves limits of** $G$ if $F(L)$ is a limiting cone to $FG$ whenever $L$ is a limiting cone to $G$.

2. $F$ **reflects limits of** $G$ if $L$ is a limiting cone to $G$ whenever $F(L)$ is a limiting cone to $FG$.

3. $F$ **lifts limits of** $G$ if every limiting cone to $FG$ is the image of some limiting cone to $G$.

4. $F$ **creates limits of** $G$ if $G$ lifts limits of $G$, and the limit is unique.

- We say $F$ **preserves/reflects/lifts/creates limits** when $F$ does for every diagram of $\mathcal{C}$.
- We say $F$ **preserves/reflects/lifts/creates small limits** when $F$ does for every small diagram of $\mathcal{C}$.

There are some properties.

- Consider the limit as $(L, \Delta_L \xrightarrow{p} G)$, an object in the comma category $(\Delta \downarrow G)$. Then $F$ preserves limits if $(F(L), \Delta_{F(L)} \xrightarrow{F \cdot p} F \circ G)$ is a terminal object in $(\Delta \downarrow F \circ G)$. Here $F \cdot p$ means the natural transformation obtained by **whiskering** $F$ **and** $p$, i.e,

$$\begin{tikzcd} \mathcal{D} \arrow[r, bend left, "\Delta_L"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow p"] & \mathcal{C} \end{tikzcd} \rightsquigarrow \begin{tikzcd} \mathcal{D} \arrow[r, bend left, "F \circ \Delta_L"] \arrow[r, bend right, "F \circ G"'] \arrow[r, phantom, "\Downarrow F \cdot p"] & \mathcal{C}' \end{tikzcd}$$

- Let $\mathcal{C}$ be a (finitely) complete category. As a consequence of Theorem 3.12, we see that a functor $F : \mathcal{C} \to \mathcal{C}'$ preserves (finite) small limits if and only if it preserves (finite) small products and equalizers.
- By Property 2.25, a functor which preserves pullbacks also preserves monomorphisms.
- Let $F : \mathcal{C} \to \mathcal{C}'$ be a limit preserving functor. If $\mathcal{C}$ is complete and $F$ reflects isomorphism, $F$ also reflects limits.
- A fully faithful functor reflects limits.

**Proposition 3.15.** Let $\mathcal{C}$ be a category and $N$ be an object of $\mathcal{C}$. Then the representable functor

$$_N h = \mathrm{Hom}_{\mathcal{C}}(N, -) : \mathcal{C} \longrightarrow (\mathbf{Set})$$

preserves all existing limits, including non-small ones. In particular, it preserves monomorphisms.

*Proof.* Consider a functor $F : \mathcal{D} \to \mathcal{C}$ with limit $(L, \Delta_L \xrightarrow{p} F)$ and a cone $(M, \Delta_M \xrightarrow{q} {}_N h(F(-)))$ in $(\mathbf{Set})$. For every element $m \in M$, the family $(q_D(m) : N \to F(D))_{D \in \mathcal{D}}$ is a cone to $F$, and therefore there's a unique morphism $q(m) \in \mathrm{Hom}_{\mathcal{C}}(N, L)$ such that for every $D \in \mathcal{D}$, $p_D \circ q(m) = q_D(m)$. Hence this defines a map $q : M \to {}_N h(L)$, with the property ${}_N h(p_D) \circ q = q_D$ for each $D \in \mathcal{D}$. The uniqueness of $q$ results from that of the $q(m)'s$. $\square$

Dually, $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(N, -) : \mathcal{C}^{\mathrm{op}} \to (\mathbf{Set})$ preserves limits, that is,

**Proposition 3.16.** Let $\mathcal{C}$ be a category and $N$ be an object of $\mathcal{C}$. Then the representable functor

$$h_N = \mathrm{Hom}_{\mathcal{C}}(-, N) : \mathcal{C} \longrightarrow (\mathbf{Set})$$

transforms existing colimits into limits and in particular epimorphisms to monomorphisms.

## 3.2 Adjoints

**Proposition 3.17.** If the functor $F : \mathcal{C} \to \mathcal{C}'$ has a left adjoint, then $F$ preserves all existing limits in $\mathcal{C}$.

*Proof.* Let $G$ be the left adjoint of $F$ and

$$\eta_{B,A} : \mathrm{Hom}(G(B), A) \longrightarrow \mathrm{Hom}(B, F(A))$$

the associated set-theoretic bijections. Consider a small diagram $H : \mathcal{D} \to \mathcal{C}$ with limit $(L, (p_D)_{D \in \mathcal{D}})$. Then $(F(L), (F(p_D))_{D \in \mathcal{D}})$ is a cone to the functor $F \circ H$. We must show $F(L)$ is the limit of $F \circ H$, and it suffices to verify the universal property.

Let $(M, (q_D)_{D \in \mathcal{D}})$ be a cone to $F \circ H$; we have $q_D \in \mathrm{Hom}_{\mathcal{C}'}(M, FH(D))$. By adjointness, $q_D$ corresponds to a unique morphism $r_D \in \mathrm{Hom}_{\mathcal{C}}(G(M), H(D))$, and by the naturality of $\eta_{B,A}$, if $d : D \to D'$ is a morphism, then $r_{D'} = {}_{G(M)}h(H(d))(r_D) = H(d) \circ r_D$.

$$\begin{tikzcd}
\mathrm{Hom}(G(M), H(D)) \arrow[r] \arrow[d, "H(d)\circ -"'] & \mathrm{Hom}(M, FH(D)) \arrow[d] \\
\mathrm{Hom}(G(M), H(D')) \arrow[r] & \mathrm{Hom}(M, FH(D'))
\end{tikzcd}$$

This means $(G(M), (r_D)_{D \in \mathcal{D}})$ is a cone to $H$, and thus there's a unique morphism $r \in \mathrm{Hom}_{\mathcal{C}}(G(M), L)$ such that $r_D = p_D \circ r$ for all $D \in \mathcal{D}$. By adjointness, $r$ corresponds to a unique morphism $s \in \mathrm{Hom}(M, F(L))$. By the naturality of $\eta_{B,A}$, we see $F(p_D) \circ s = q_D$.

$$\begin{tikzcd}
\mathrm{Hom}(G(M), L) \arrow[r] \arrow[d, "p_D\circ -"'] & \mathrm{Hom}(M, F(L)) \arrow[d, "F(p_D)\circ -"] \\
\mathrm{Hom}(G(M), H(D)) \arrow[r] & \mathrm{Hom}(M, FH(D))
\end{tikzcd}$$

Since $\eta_{B,A}$ is bijective, $s$ is unique with this property. $\square$

**Proposition 3.18.** A category $\mathcal{C}$ is complete (resp. cocomplete) if and only if for every small category $\mathcal{D}$, the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C}^{\mathcal{D}}$ has a right (resp. left) adjoint.

*Proof.* This follows from Example 3.3 □

## The adjoint functor theorem

**Theorem 3.19.** Let $\mathcal{D}$ be a complete category. Then $\mathcal{D}$ has an initial object if and only if it satisfies the **solution set condition**, that is, there are a set $I$ and an $I$-indexed family $(S_i)_{i\in I}$ of objects of $\mathcal{D}$ such that for every $D \in \mathcal{D}$ there's an arrow $S_i \to D$ for some $i \in I$.

*Proof.* The necessity is clear; if $\mathcal{D}$ has an initial object $S$, then $S$ indexed by a one point set satisfies the solution set condition. Conversely, we assume the solution set condition. Since $\mathcal{D}$ is complete, $P := \prod_{i\in I} S_i$ exists. Since $\mathrm{End}_{\mathcal{D}}(P)$ is a set, the equalizer $e : E \to P$ of all endomorphisms of $P$ exists. For each $D \in \mathcal{D}$, there's at lest one arrow $E \to D$, given by $E \xrightarrow{e} P \xrightarrow{p_i} P_i \to D$ for some $i \in I$. Suppose there are two morphisms $f, g : E \to D$, and consider their equalizer $E' \xrightarrow{u} E$. Then for some $j \in I$, we have the composite of morphisms $P \xrightarrow{p_j} S_j \xrightarrow{h} E' \xrightarrow{u} E \xrightarrow{e} P$. Since $e$ is the equalizer of $\mathrm{End}_{\mathcal{D}}(P)$, it gives

$$(euhp_j)e = \mathrm{id}_P\, e = e\, \mathrm{id}_E$$

Being an equalizer, $e$ is a monomorphism, and thus $uhp_je = \mathrm{id}_E$, meaning that $u$ has a right inverse. Then $u$, being a monomorphism having a right inverse, is an isomorphism, and thus $f = g$. □

**Theorem 3.20** (The Freyd Adjoint Functor theorem)**.** Let $\mathcal{D}$ be a complete category. Let $F : \mathcal{D} \to \mathcal{C}$ be a functor. TFAE:

1. $F$ has a left adjoint.
2. $F$ satisfies the following two conditions:
   (a) $F$ preserves small limits.
   (b) the **solution set condition** holds, that is, for each object $N \in \mathcal{C}$, there are a set $I$ and an $I$-indexed family of morphisms $f_i : N \to F(A_i)$ such that every morphism $h : N \to F(A)$ can be written as a composite $h = F(t) \circ f_i$ for some $i \in I$ and $t : A_i \to A$.

*Proof.* If $F$ has a left adjoint $G$, then it preserves all limits by Proposition 3.17 and the universal arrows from $N$ to $G$ satisfy the solution set condition, with $I$ being the one-point set.

Conversely, to say $F$ admits a left adjoint, it suffices to construct a universal arrow from $N$ to $F$ for each $N \in \mathcal{C}$. By Proposition 2.5, it suffices to show the comma category $(N \downarrow F)$ admits an initial object. In view of Theorem 3.19, it suffices to show $(N \downarrow F)$ is complete and satisfies the solution set condition; the latter is clear by assumption. To show the former, it suffices to show $(N \downarrow F)$ admits small products and equalizers, that is,

**Lemma 3.21.** If $F : \mathcal{D} \to \mathcal{C}$ preserves all small products (resp. all equalizers), then for each $N \in \mathcal{C}$, the projection

$$Q : (N \downarrow F) \to \mathcal{D},\ (N \xrightarrow{s} F(A)) \mapsto A$$

of the comma category creates all small products (resp. all equalizers).

*Proof.* Let $J$ be a set and $f_j : N \to F(A_j)$ be a $J$-indexed family of objects of $(N \downarrow F)$ such that the product $\prod_j A_j$ exists in $\mathcal{D}$. Since $F$ preserves products, $F(\prod_j A_j)$ is a product of $F(A_j)$ in $\mathcal{C}$. Thus there's a unique morphism $f : N \to F(\prod_j A_j)$ such that $f_j = F(p_j) \circ f$, where the $p_j : \prod_j A_j \to A_j$ are the projections. One can show that $(f, G(p_j))$ is the product of $f_j$ in $(N \downarrow F)$, and thus $Q$ reflects products.

Let $s, t : f \to g$ be a parallel of morphisms in $(N \downarrow F)$, where $f : N \to F(A)$, $g : N \to F(B)$. Consider the equalizer $e : C \to A$ of $s, t$ in $\mathcal{D}$. Since $F$ preserves equalizers, $F(e) : F(C) \to A$ is a equalizer of $F(s), F(t)$ in $\mathcal{D}$. Since $F(s) \circ f = g = F(t) \circ f$, there's a unique morphism $h : N \to F(c)$ such that $f = F(e) \circ h$. If there are $u : N \to F(D)$ and $v : D \to A$ such that $s \circ v = u \circ v$ in $(N \downarrow F)$, then $s \circ v = u \circ v$ in $\mathcal{D}$, so there's a unique morphism $r' : D \to C$ such that $e \circ r' = v$. We show $r'$ is a morphism in $(N \downarrow F)$; indeed, $F(e) \circ (F(r') \circ u) = F(v) \circ u = f$, so by the uniqueness of $h$, we have $h = F(r') \circ u$, implying that $r'$ is a morphism in $(N \downarrow F)$. In conclusion, $e$ is the equalizer of $s, t$ in $(N \downarrow F)$.

□

**Theorem 3.22.** Let $\mathcal{D}$ be a complete category. A functor $F : \mathcal{D} \to (\mathbf{Set})$ is representable if and only if $F$ preserves all small limits and satisfies the **solution set condition**, that is, there is a set $\mathcal{S}$ of objects of $\mathcal{D}$ such that for any object $D \in \mathcal{D}$ and any element $A \in F(D)$ there exist an $S \in \mathcal{S}$, a $B \in F(\mathcal{S})$ and an arrow $f : S \to D$ such that $K(f)(B) = A$.

*Proof.* A representative of $F$ is an initial object in the comma category $(* \downarrow F)$, by Corollary 2.5.1. Then the if part follows by Theorem 3.19. The only part follows by Proposition 3.15. □

**Example 3.23.** Let $U : (\mathbf{Grp}) \to (\mathbf{Set})$ be the forgetful functor. $(\mathbf{Grp})$ is complete and $U$ clearly preserves all small-limits. For each set $X$, consider any map $f : X \to U(G)$ for some group $G$ and let $S = \langle f(X) \rangle \leqslant G$.